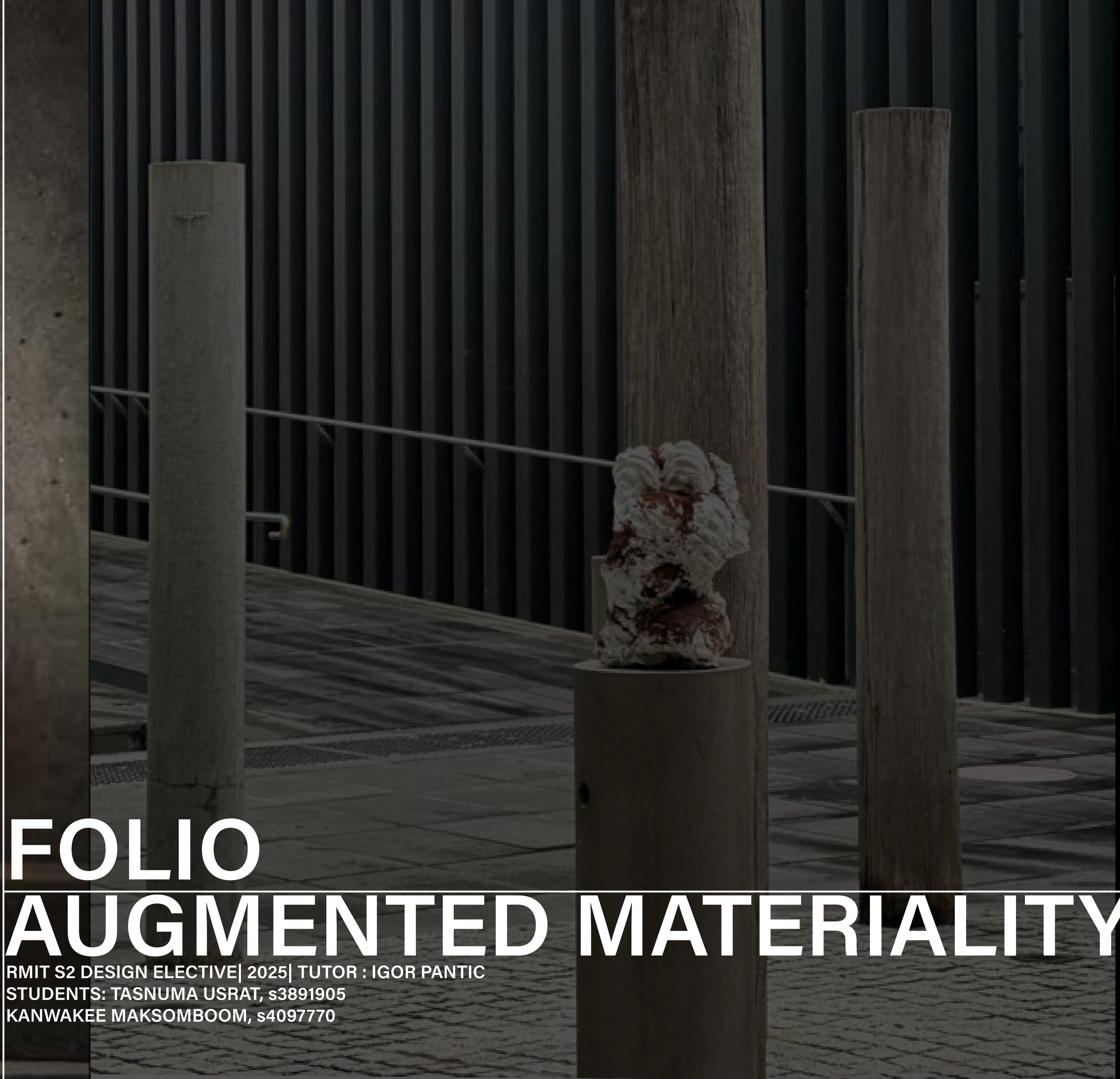


FOLIO
AUGMENTED MATERIALITY

RMIT S2 DESIGN ELECTIVE| 2025| TUTOR : IGOR PANTIC

STUDENTS: TASNUMA USRAT, s3891905
KANWAKEE MAKSOMBOOM, s4097770

CONTENTS

MATERIAL RESEARCH	2-15
FORMING STRATEGY	16-17
MATERIAL TESTING	18-29
GEOMETRY STUDY	30-43
CASTING/CARVING TESTS	44-73

MATERIAL RESEARCH : USED COFFEE GROUNDS

The primary material for this research is a high-volume waste product that is easily accessible from homes or cafés, particularly within Melbourne's urban context. The aim is to explore the potential of everyday waste as a sustainable construction or design material, reducing landfill impact and promoting circular design practices.

CORE QUALITIES :

Strength Enhancement:

Coffee biochar, made by roasting coffee grounds, increases the 30% strength and durability of concrete. It also reduces the need for sand and cement, lowering costs and environmental impact.

Environmental Benefits:

Reduces landfill waste by diverting coffee grounds.
Decreases reliance on finite resources like mined sand.
Lowers the carbon footprint of concrete production.

Real-World Applications:

Trials are active in Melbourne, including a **footpath and a road project.**

Researchers have also developed **sustainable bricks** from coffee waste, baked at lower temperatures.

Ongoing Research:

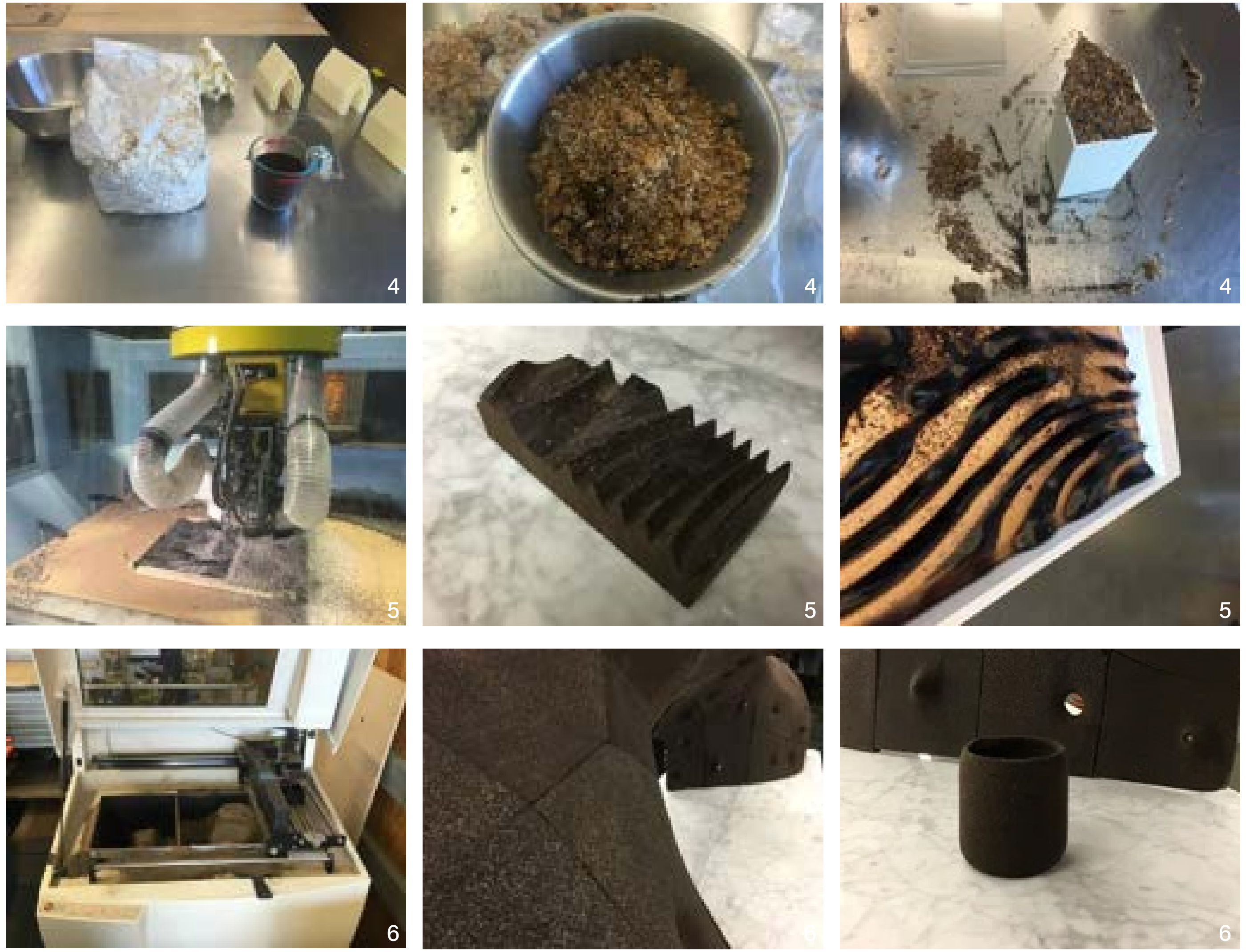
RMIT University is exploring the ideal ratios of coffee grounds, sand, and **cement for maximum performance.**

Research is also looking into how coffee biochar can improve **concrete's thermal and acoustic properties.**

Source : https://www.rmit.edu.au/news/all-news/2023/aug/coffee-concrete

CRAFTING/ FORMING STRATEGY

There are three distinct fabrication methods, each yielding different material qualities and geometric possibilities



Source : https://www.instructables.com/How-to-Reuse-Your-Old-Coffee-Grounds-for-Fabricati/

FABRICATION METHODS :

1. Sourcing Coffee Grounds

Get grounds from home or local cafes. It is widely available as waste material, coffee grounds are the second most traded commodity. Their porous nature makes them prone to mold if not collected or stored properly.

2. Processing Coffee Grounds

Collected grounds are typically wet and must be dried before use.

Quickly dry grounds in an oven at 400°F (200°C) for 15-20 minutes, watching carefully to prevent burning.

Alternatively, spread them in the sun to dry naturally.

3. Fabrication Possibilities

Coffee grounds can be fabricated into materials with various qualities depending on the process.

This material can be used with Mycelium, polymer composites, or 3D printing, each creating a unique material.

4. Coffee for Growing Mycelium

Crafting Strategy: Used coffee grounds provide nutrients for mycelium to grow and bind materials together.

Process: Mix sterilized coffee grounds with a mycelium starter, place in a mold, and let it grow for 5-8 days in a warm, dark, humid environment.

Geometric Possibilities: Creates objects that conform to the formwork or molds. The final product is a **solid, insulative form suitable for biomass infill** or unique architectural components.

5. Coffee for Composite Polymers

Crafting Strategy: Coffee grounds act as a **biomass (sand) filler or raw material** for casting with resins like epoxy or polyester.

Process: The correct ratio of coffee to resin creates desired **translucency, texture, and strength.**

Geometric Possibilities: This material works well for architectural **cladding, panels, and feature walls.**

6. Coffee for 3D Printing

Crafting Strategy: Used coffee grounds can be **powder printed** if ground to a fine consistency (**below 50 microns**).

Process: A binder solution is applied layer by **layer to the coffee powder, solidifying** it into a three-dimensional object.

Geometric Possibilities: The process is ideal for creating **unique, intricate objects and non-standard parts.** like architectural **facades or tiles.**

COFFEE WASTE TURNS INTO 3D PRINTED FURNITURE

The project explores several geometric intents, combining fluid forms with modular assemblies.

Fluid / Sculptural Forms:

The **overall design** is inspired by the "fluidity of coffee." The counters and stools feature soft, undulating, and organic surfaces. **The stool design** in particular mimics the shape of a coffee bean, translating a small, natural form into a large-scale, functional object.

Layered Surfaces:

The 3D printing process naturally **creates a linear / ruled logic** through its horizontal layers. The visible layer lines and color variations (caused by different densities of coffee grounds) become a defining aesthetic feature, especially when backlit, which enhances their unique texture.

Modular Elements:

While **the pieces are custom and unique**, the project demonstrates how an **entire space can be defined by a series of custom, parametric-designed** elements that are geometrically coherent and function as a system. The counter, lamps, and stools work together to **create a unified design.**

Source : https://www.instructables.com/How-to-Reuse-Your-Old-Coffee-Grounds-for-Fabricati/

VISUAL REFERENCES





POSSIBLE WAYS MR/AR CAN ASSIST:

MR/AR system could be integrated to enhance the process, particularly for live, freeform adjustments or on-site assembly. This would be a "Human-in-the-loop alternative to automation" or an **"Augmenting an existing method"** approach.

Real-Time Design Adjustment:

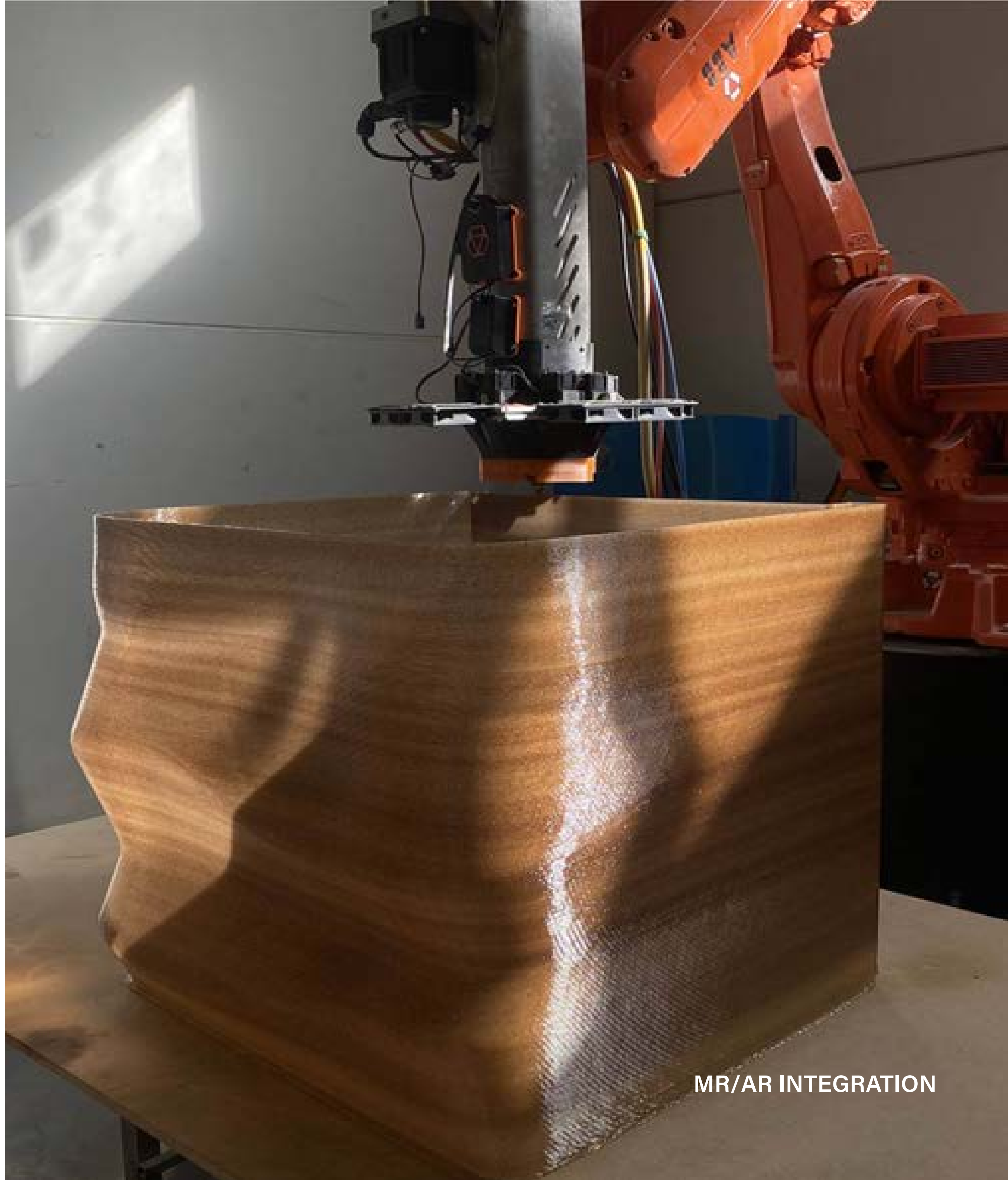
The AR headset could display a live "ghost" of **the digital model projected directly onto the physical print in progress.** This allows the operator to perform real-time quality control and make on-the-fly design modifications. For example, **the AR overlay could guide the operator to manually smooth a surface or add a texture with a hand tool as the material is extruded.**

Precision Tool Guidance:

For post-processing tasks like CNC milling or sanding, the AR system could **highlight specific areas that need work**, ensuring the final finish matches the design intent perfectly.

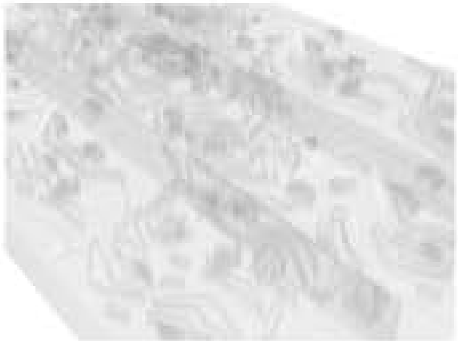
Live Sensor Feedback:

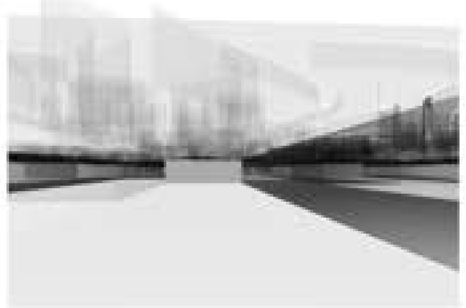
The headset could display live data from sensors on the robotic arm, such as **material temperature, extrusion speed, and layer height.** This would allow the human operator to catch errors immediately, improving the final product's quality and reducing material waste.


MR/AR INTEGRATION

MATERIAL RESEARCH : TRANSLUCENT WOOD

A translucent wood is crafted by removing lignin and filling the pores of the wood fibres with clear resin.





CORE QUALITIES :

- Renewable wood with lower energy input
- Lightweight and stronger than traditional wood
- 90 % light transmittance
- Warm- diffused lighting
- Reduced flammability compared to untreated wood
- Anisotropic Optics

Lightweight but Strong for Its Weight

- Density: 300–500 kg/m3, lighter than glass (2,500 kg/m3).
- Tensile strength and toughness remain higher than plain glass due to retained cellulose nanofibres.
- Can be used for structural panels, skylights, or partitioning with reduced weight load.

Optical Properties

- High light transmittance: 80–90% when optimised, while maintaining wood's fibrous texture.
- Diffuses light instead of allowing direct glare, providing softer illumination indoors.
- Can be engineered for varying transparency (semi-opaque to nearly clear).

Thermal + Acoustic Insulation

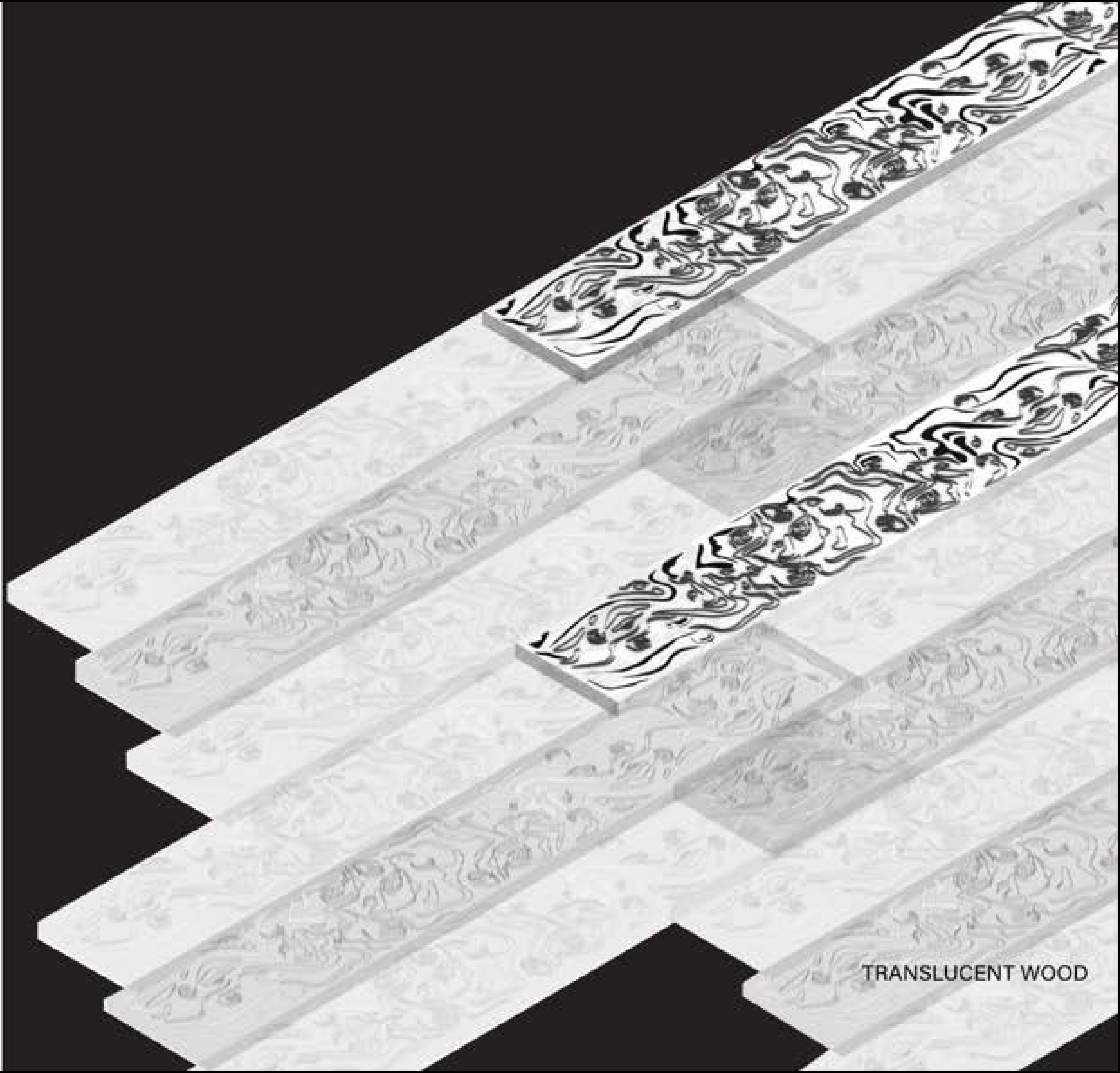
- Thermal conductivity: 0.3–0.4 W/mK (lower than glass, offering better insulation).
- Maintains porous wood structure for partial sound absorption, useful in interior walls.
- Energy-saving potential for sustainable architecture.

Fire Resistance

- By integrating polymers with treated cellulose, it chars rather than shattering dangerously like glass.
- Can be enhanced with fire-retardant resins for interior applications

Biodegradable + Sustainable

- Derived from renewable timber sources instead of petroleum-based plastics.
- Maintains carbon storage from natural wood growth.
- Potentially recyclable or reprocessable, depending on polymer used.



TRANSLUCENT WOOD

CRAFTING/ FORMING STRATEGY

FABRICATION METHOD :

Translucent wood is typically fabricated through adaptations of traditional woodcraft combined with polymer infiltration.
The following techniques are most relevant:

Laminating: Thin delignified veneers are stacked and bonded with transparent polymer to increase thickness and mechanical strength.

Bending / Controlled Deformation: Veneers can be softened during resin impregnation and pressed into curved molds before curing.

Casting: Resin infiltration itself functions like a casting process — polymer flows into the natural wood cellular structure, hardening in place.

Panel Assembly: Multiple sheets are pressed into panels, sometimes hybridised with glass, acrylic, or biopolymer films for structural reinforcement.

Stacking / Tiling: Smaller translucent wood units can be tiled into larger surfaces, introducing modularity and reducing waste.

Process Execution :

Delignification

Thin wood slices (1–5 mm) placed in sodium chlorite bath at 70–80 °C.

Output: white, porous cellulose framework.

Resin Impregnation

Wood slices submerged in transparent polymer under vacuum/pressure.

Polymer replaces lignin, improving transparency and strength.

Curing

Heat or UV curing solidifies resin in situ.

The composite becomes transparent but retains natural fibre texture.

Forming Techniques :

Laminating: Veneers stacked into thicker sheets.

Bending: Panels pressed into curved molds before resin sets.

Panel Assembly: Larger modules joined with adhesives or hybrid lamination.

Finishing: Sanding/for optical clarity. Protective UV coatings applied.

Potential Inefficiencies / Areas for Improvement

Energy & Chemical Use:
Delignification requires high heat and chemical solutions. Sustainable alternatives (enzymatic or green oxidisers) could reduce environmental footprint.

Resin Choice:
Current methods rely on petroleum-derived epoxies or PMMA. Research into bio-based, biodegradable resins would align with the ecological ethos.

Thickness Limitations:
Transparency drops with thicker panels due to light scattering. Layering strategies or hybrid laminates (wood + acrylic) may overcome this.

Surface Yellowing:
UV exposure causes yellowing over time. Protective films or embedded stabilisers are required for longevity.

GEOMETRY INTENT/ VISUAL REFERENCES

Translucent wood as a futuristic material carries the potential to merge the warmth of natural grain with the ethereal qualities of glass, producing spaces that feel both grounded and otherworldly. Its design intent lies in using light as a material in itself — filtering, diffusing, and transforming daylight into soft amber glows that blur the line between opacity and transparency.

In the **Philharmonic Hall** in Szczecin by Barozzi/Veiga, the rhythmic white ridges of the façade create a monumental civic presence through sharp geometry and light play; reimagined in translucent wood, such vertical striations would radiate a softer, honey-coloured luminescence, offering a monumentality that feels humanised and tactile rather than cold.

Similarly, the **TeaHouse** in Copenhagen by Pan Projects employs a timber lattice to frame ritual and intimacy; if reinterpreted with translucent wood, the pavilion would become a glowing lantern-like enclosure, diffusing golden gradients of light that shift with time of day and orientation.

Geometrically, translucent veneers could be laminated into rhythmic panels, bent into curved funnels, or tiled into modular skins, but always with the effect of turning surface into atmosphere. Environments shaped with this material would feel calm and contemplative by day, filtering natural light without severing privacy, and at night they would transform into luminous beacons, glowing from within like civic lanterns.
Unlike glass, which conveys cold transparency, or concrete, which asserts heavy opacity, translucent wood produces an affective middle ground — an architecture of gentle opacity and visible life that invites both exhibition and retreat, creating an atmosphere of biophilic futurism.

Philharmonic Hall by Barozzi / Veiga, Szczecin, Poland

Teahouse by PAN-PROJECTS, Copenhagen, Denmark

POSSIBLE WAYS MR/AR CAN ASSIST:
When working with translucent wood as an architectural surface, the focus is on how light is curated through joints, layers, and surface finishes. MR/AR can be used as a design and fabrication tool to guide the placement of panels, alignment of timber joints, and treatment of surfaces so that the intended lighting effect is achieved.

Real-Time Light Studies:
AR can project simulated daylight or artificial light conditions directly onto the translucent panels as they are placed. This lets the designer see how light will filter through, revealing where adjustments to panel thickness, grain direction, or joint spacing are needed to refine the atmosphere of the interior.

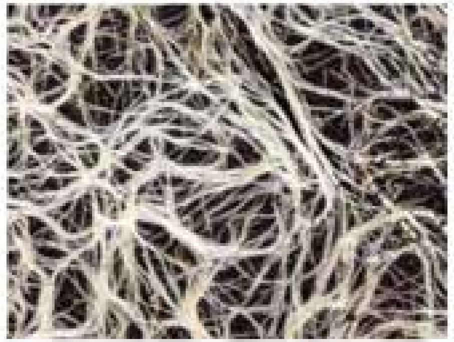
Guided Joinery and Panel Alignment:
Timber joints can disrupt or enhance the light quality depending on how they are cut and assembled. AR overlays can guide precise alignment of dovetails, lap joints, or laminated edges so that seams read as part of the architectural joinery rather than visual interruptions.

Material Performance Feedback:
Sensors can track curing of resin within the translucent wood, moisture content in the timber, or stress along structural seams. MR displays this live information, allowing the maker to intervene before irregularities distort the light quality or weaken the structure.

Through the integration of MR/AR, the use of translucent wood shifts from a material experiment to a carefully choreographed architectural medium. Light is no longer incidental but becomes a designed element, directed through the grain, joints, and layered finishes of the timber. By projecting lighting conditions in real time and guiding fabrication with precision, we can extensively curate the interplay of light- soft diffusion, sharp contrasts, or glowing seams—transforming translucent wood into an active surface that shapes atmosphere and spatial experience.

MATERIAL RESEARCH: MYCELIUM

Mycelium is the rooty, fibrous network of fungi that transforms organic waste into solid, sculptable, and eco-friendly material.



CORE QUALITIES :

1. Lightweight but srong for its weight

Mycelium composites are low density like a foam (~40–200 kg/m³).

Strong enough for non-load-bearing panels, insulation, or furniture

Compressive strength is around ~0.1–0.2 MPa (compared to concrete = 17–28 MPa).

2. Insulation

Thermal conductivity: ~0.04–0.06 W/mK, similar to polystyrene insulation.

Acoustic absorption: it has porous structure that absorbs sound, useful for interior cladding.

3. Fire resistantance

Unlike plastics/foams, mycelium chars instead of dripping toxic flames.

Can improve fire safety in interiors compared to petrochemical foams.

4. Biodegradable + Sustainable

Made from agricultural waste + fungi.

Fully compostable at end-of-life.

Functions as a carbon sink while growing (locks CO_2 into its structure).






FABRICATION METHOD

1. Substrate Preparation

Start with an organic waste substrate (e.g. sawdust, hemp hurds, straw, cardboard, or agricultural residue).

Clean/sterilize it (usually steam-sterilized) so competing organisms don't outgrow your fungus.

2. Inoculation

Mix the substrate with mycelium spawn (the fungal "starter culture").

Common fungi used: Ganoderma lucidum, Pleurotus ostreatus (oyster), or Ganoderma applanatum.

This is usually done in a clean, semi-sterile environment to avoid contamination.

3. Growth Phase

Place inoculated substrate in molds (bricks, panels, curved shells, etc.).

Conditions: ~20–28 °C temperature, ~70–90% humidity, darkness, and oxygen supply.

Mycelium digests the substrate, binding it into a solid, foam-like composite. Growth takes about 4–10 days depending on thickness.

4. Forming / Shaping

Mold Method: Substrate is placed in a rigid mold (like a brick or a curved shell). Mycelium grows to fill it.

Freeform Method: Combine with reinforcement (like textiles, timber lattice, or 3D-printed skeletons), then let it colonize around the frame.

Layering: Press substrate/mycelium mix into thin sheets. Used for panels.

5. Deactivation (Drying & Heat Treatment)

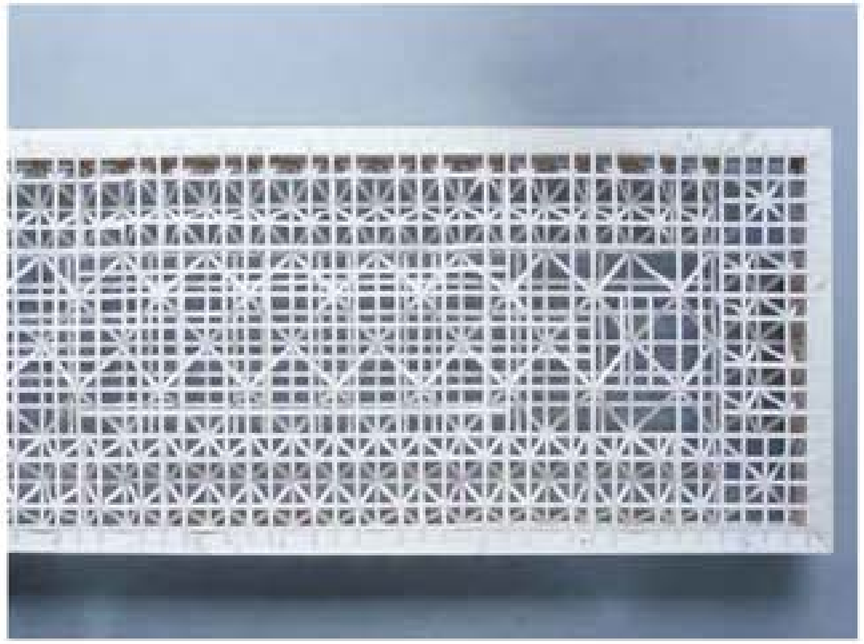
Once the desired form is achieved, the material is baked or dried at ~80–100 °C.

This halts growth (so the model doesn't sprout mushrooms later), reduces moisture, and stabilizes the structure.

6. Post-Processing

Surface finishing: sanding, cutting, CNC milling, or coating (biodegradable sealants, resins, or wax).

Structural enhancement: some labs test hybrid composites (e.g. mixing with natural fibers, or sandwiching mycelium panels in timber/metal frames).



A WOOD-MYCELIUM COMPOSITE CONSTRUCTION

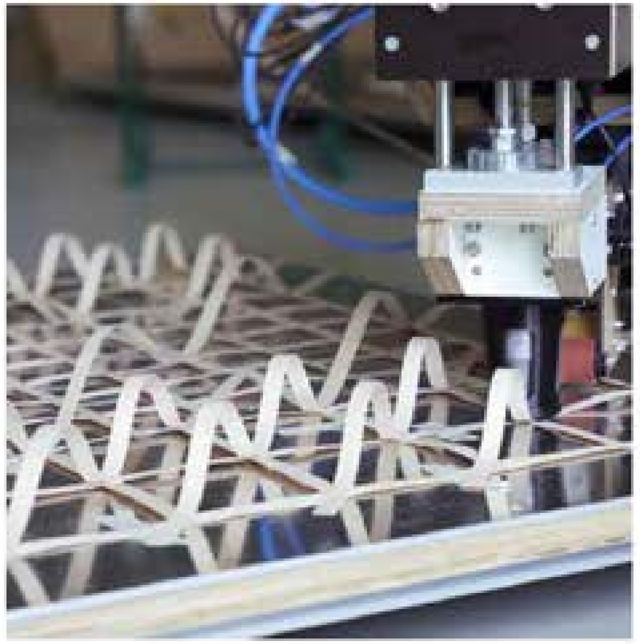
The researchers in University of Kassel in Germany developed an **automated glueless process** to efficiently create 3D lattice structures from indigenous **maple wood**. This wood serves as reinforcement and moulding scaffolds for the mycelium to **grown on**.

The wood ensures the composite's load-bearing capacities.

The aim of the project was to create **CO2 neutral**, circular fittings to upgrade existing office buildings.

University of Kassel

Source: https://materialdistrict.com/article/a-wood-mycelium-composite-construction/



MYCELIUM SUPPORTED ON A TIMBER FRAME

The Growing Pavilion is a temporary events space at Dutch Design Week constructed with panels grown from mushroom mycelium supported on a timber frame.

Designed by set designer and artist Pascal Leboucq in collaboration with Erik Klarenbeek's studio Krown Design at Amsterdam studio Biobased Creations, the temporary pavilion is made entirely from **bio-based** materials.

The outer panels were **grown from mushrooms**, with the mycelium in the roots providing strength. These are covered with a coating that is a bio-based product originally developed by the Maya people in Mexico.

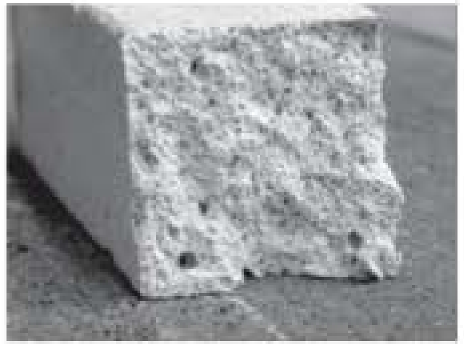
The panels were attached to a **timber frame**, and can be removed and repurposed as necessary. The floors are made from cattail – a type of reed – with interior and exterior benches made from agricultural waste.

Source: https://www.dezeen.com/2019/10/29/growing-pavilion-mycelium-dutch-design-week/

MATERIAL RESEARCH

CELLULOSE–GYPSUM COMPOSITES

Gypsum composites are materials composed primarily of gypsum, a soft sulfate mineral, combined with various additives to enhance its properties. These composites are widely used in construction due to their fire resistance, sound insulation, and mold resistance characteristics. They are commonly employed in interior walls, ceilings, and partitions.



CORE QUALITIES :

Overview

- Composite of gypsum plaster, reinforced with fibres (recycled paper pulp, jute/hessian, or fiberglass mesh).
- Castable, fast-setting, low-toxicity; highly repeatable in simple molds.

Mechanical & Physical

- High compressive strength; brittle in tension + mitigate with mesh/ pulp and back ribs.
- Good surface detail; can be thin (10–20 mm) with local thickening at edges/ribs.
- Thermal stability and good fire behaviour

Dimensional Stability

- Low chemical shrinkage;
- Practical tolerances ±0.5–1.0 mm with consistent molds and QA.

Moisture & Durability

- Hygroscopic when unsealed; edges need sealing in humid areas.
- Repairable: chips can be filled and sanded; tiles are easy to replace individually

Surface / Finish

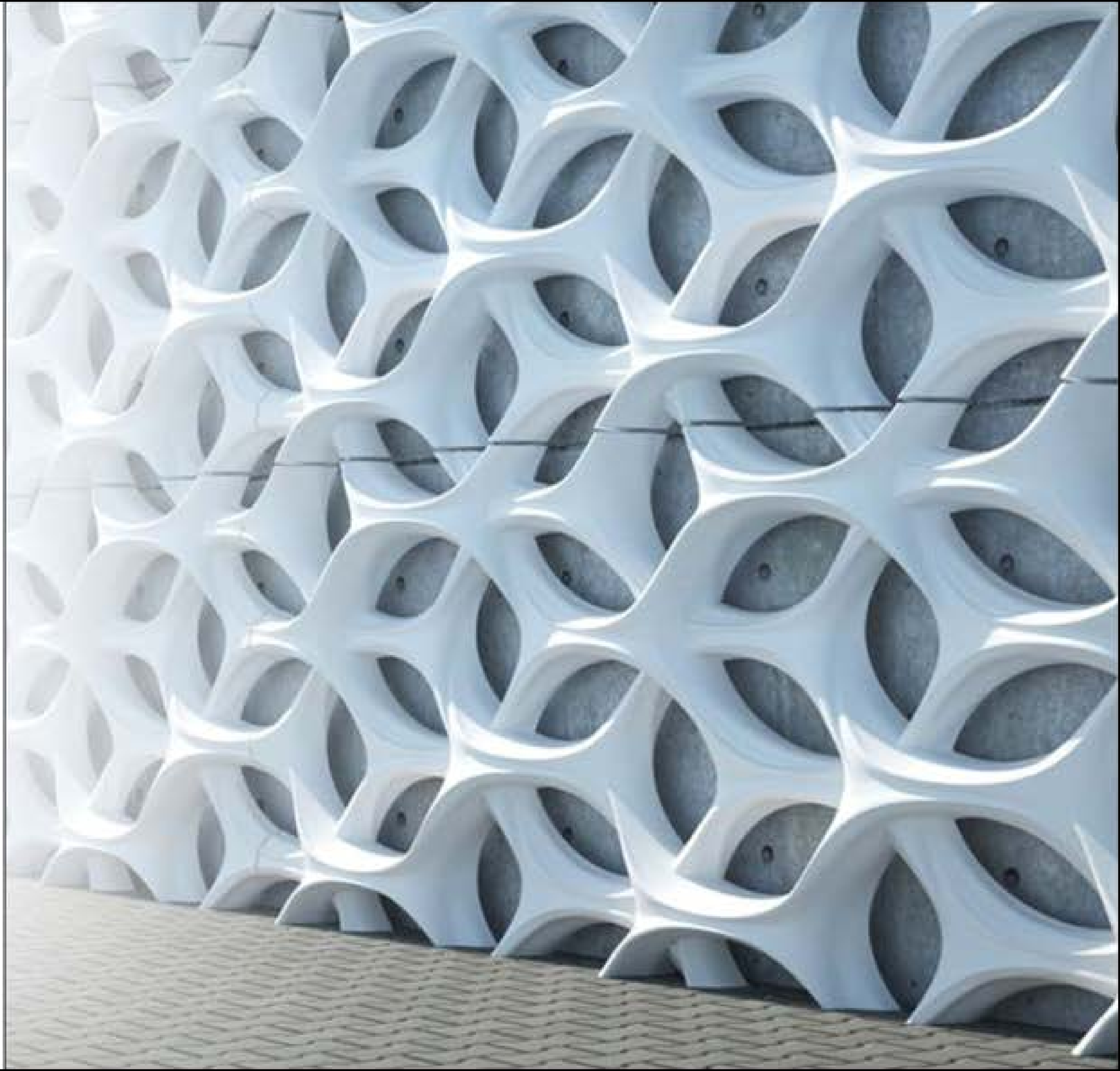
- Accepts pigments, stains, paints; takes liner textures (fabric/mesh) and inlays (glass/resin lenses).
- Sandable, drillable; compatible with embedded inserts (M4 brass) or magnets.

Environmental Profile

- Low embodied energy vs Portland cement; can include recycled cellulose fibres.
- Waste/rejects can be crushed and reused as filler (non-structural).
- Low-VOC; dust control needed during sanding.

Applications

- Interior cladding tiles, acoustic relief panels, coffers, ribs; prototyping of façade systems.
- Distributed casting compatible with community or multi-team production.



Gypsum-reinforced composites	Material properties	Weaknesses	Applications
Glass Fiber Reinforced Gypsum (GFRG)	GFRG is known for its high strength and low weight, making it suitable for construction applications where structural strength is needed. It's fire-resistant and offers good insulation properties	The raw materials can be expensive, and manufacturing GFRG panels can be complex and require specialized machinery	Load-bearing structures, walls, partitions
Glass Fiber Reinforced Cement (GFRC) with Gypsum	GFRC with gypsum combines the durability of cement with the lightweight properties of gypsum. It has good resistance to weathering and is used in decorative and architectural applications	The manufacturing process is labor-intensive and can be expensive, making it more suitable for high-end applications	Architectural facades, decorative elements, custom molding
Natural Fiber Reinforced Gypsum (NFRG)	It has lower mechanical strength compared to other types, limiting its use in load-bearing applications	It has lower mechanical strength compared to other types, limiting its use in load-bearing applications	Ceiling tiles, decorative panels, sustainable construction
Synthetic Fiber Reinforced Gypsum	These composites can be tailored for specific properties, including high strength, impact resistance, and fire resistance. They are versatile and suitable for various applications	Depending on the synthetic fibers used, cost and availability can vary. Manufacturing can also be complex, especially for advanced composites	Fire-resistant boards, durable panels
Recycled Gypsum Composites	These composites are eco-friendly, as they utilize recycled gypsum waste. They offer good insulation properties and are relatively cost-effective	Mechanical properties can be inferior to other types, limiting their use in high-stress applications	Wallboards, non-structural elements, interior finishes
Exterior Gypsum Sheathing	This type is specifically designed for exterior sheathing applications. It provides weather resistance and good moisture control, helping protect the building envelope	Primarily suited for specific construction needs, not suitable for decorative or load-bearing applications	Exterior sheathing in residential and commercial buildings

FABRICATION METHODS

Co-calcining and layering:
This method involves heating gypsum and cellulose fibers together to create a composite slurry, which is then layered onto a forming screen. Multiple layers can be created, including fiber layers and gypsum layers, often using different headboxes for each.

Reinforcing mesh:
A reinforcing mesh, often made of a material like fiberglass, can be incorporated into the gypsum-fiber mixture during the forming process. The mesh is embedded within the gypsum matrix, improving the composite's impact resistance and overall strength.

Compression molding:
In this method, the gypsum and fiber mixture is compressed into a mold to create a specific shape, like a board or panel.

Fiber pretreatment:
Natural cellulose fibers, like flax or hemp, may require pretreatment to improve their compatibility with the gypsum matrix and reduce water absorption.

Adding cellulose ether:
Cellulose ether additives can be incorporated into the gypsum slurry to enhance properties like nail pull resistance and flexural strength.

Multilayer construction:
The composite can be constructed with multiple layers, such as a core layer of gypsum and cellulose fiber and outer layers of cellulose fibers, to further optimize properties like surface finish and strength.

Considerations:

Fiber type and properties:
The type of cellulose fiber (natural or synthetic, length, etc.) significantly impacts the final composite's properties.

Fiber content:
The ratio of fibers to gypsum in the composite mixture influences strength, flexibility, and other propertie.

MR/AR INTEGRATION
Possible ways MR/AR can assist

Human-in-the-Loop Alternative to Automation

- Mold Prep (Indirect MR): Project tile outline, edge key offsets, and insert pads to standardise setup.
- Casting Assist (Indirect MR): Live pour height bands, timers for set window, warnings for premature demold.

Augmenting an Existing Method (Direct MR)

- Assembly Layout: Display grid, tile IDs, and tolerance halos; halo turns green at ±1 mm fit; show magnet polarity/screw locations.
- Quality Control: Pass/fail overlays for edge gaps, flushness, and cumulative drift after each row.

Expected Benefits

- Reduced rejects; faster cycle time; consistent connector placement; measurable tolerance control across multi-tile assemblies.

Toolchain

- Rhino/Grasshopper + Fologram; Meta Quest / phone AR.

GEOMETRY INTENT/ VISUAL REFERENCES

MATERIAL TESTING

| Introduction: Bo-base material approaches using waste coffee grounds as the strater,and bio miture glues

BASE MATERIAL :
WASTE COFFEE GROUNDs

Drying

Toasting : 4-5 mins
or Baking : 45 mins

Resting

30 mins

DRIED COFFEE GROUND

TYPE A : SOLID COFFEE
Pure bio-material | Solid wasted cofe ground composite





water + gelatin + syrup + cooking oil + coffee ground

TYPE B : TRANSLUCENT COFFEE
Pure bio-material | Translucent | Thin | Flixible





water + starch + glycerine + vernegar + coffee ground

MATERIAL TESTING

Overview: Approaches Using Waste Coffee Grounds

Material criteria

Physical strength
Remaining shape
Castability
Curing duration
Material finish

MATERIAL TESTING

| Type A : Ingredient ratio test

TYPE AA

water : gelatin : syrup : cooking oil : coffee ground
10 : 1.5 : 1.5 : 0.1 : 3

Takes longer to dry, but remains controllable
Note: Rougher surface compared with Type AB



TYPE AB

water : gelatin : syrup : cooking oil : coffee ground
10 : 3 : 1.5 : 0.1 : 3

Ideal ratio, proper shape retention, fast drying, smooth surface
Note: Requires a quick casting process



TYPE AC

water : starch : vinegar : coffee ground
10 : 3 : 2 : 3

Using cornstarch as a substitute for gelatin — a cheaper option, resulting in a rough and matte finish
Note: Breakable when casting in a non-stiff mold



Result

Strength
AA ●●○
AB ●●●
AC ●○○

Casting ability
AA ●●○
AB ●●●
AC ●●●

Transparency
AA ●●○
AB ●●○
AC1 ●○○

Smoothness
AA ●●○
AB ●●●
AC ●○○

Surface dry
AA 2 Hr.
AB 1 Hr.
AC 30 min

Material dry
AA 1 week
AB 1 week
AC N/A

MATERIAL TESTING

| Type A : Ingredient ratio test

TYPE AD
PVA : Constarch water ratio (1:4) : coffee ground
3 : 1 : 3

Medium strength, Mid-cure 6 Hr.
Note: Adding constarch mixture for liquid mixing purpose
,while the material break when demold



TYPE AE
Quick-Dry PVA : Constarch water ratio (1:4) : Coffee ground
3 : 1 : 3

Medium strength, Mid-cure 4 Hr.
Note: Adding constarch mixture for liquid mixing
purpose, while constarch effect material strength



TYPE AF
PVA : coffee ground
1 : 1

Medium strength, Mid-cure 4 Hr.
Note: Mid-cure can result in cracking.



TYPE AG
Quick-Dry PVA : Coffee ground
1 : 1

High strength, Mid-cure 2 Hr.
Matt finish, Pefect for casting
Note: Edge condition



MATERIAL TESTING

| Type A : Ingredient ratio test

TYPE AH
Polyurethane : coffee ground
1 : 1

Retain shape, stiff & high strength, fast cure 1 hr.
High gloss finish, Pefect for casting
Note: Expandable size when open casting. Suitable for pressed mold

Glue behaviour
Polyurethane : PVA : Quick-Dry PVA









MATERIAL TESTING

|Type A process

MATERIAL TESTING

| Paper mold casting : Type AB

MATERIAL TESTING

| Geometry casting : Type AB

WASTED COFFEE GROUNDS

MATERIAL TESTING

MATERIAL TESTING

| Type B : Ingredient ratio test

TYPE BA

water : starch : vinegar : glycerin
10 : 1.5 : 1 : 1

Higher strength and fast drying

Note: The material itself requires a long time to set

TYPE AC

water : starch : vinegar : glycerin : coffee ground
10 : 1.5 : 1 : 1 : 0.5

Translucent sparkling coffee effect

Note: Coffee grounds may reduce the strength of the material
The material itself requires a long time to set

Result

Strength
BA ●●●
BB ●●○

Smoothness
BA ●●●
BB ●●●

Casting ability
BA ●●○
BB ●●○

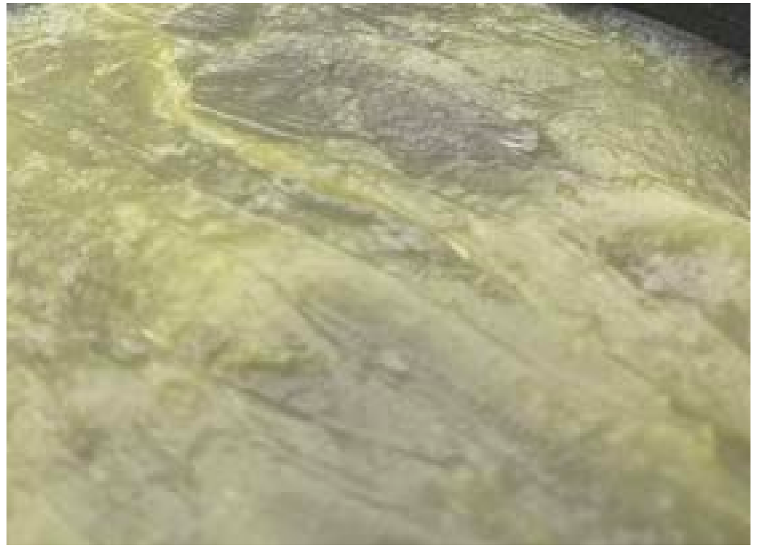
Surface dry
BA 1 Hr.
BB N/A.

Transparency
BA ●●●
BB ●●○

Material dry
BA 1 week
BB 1 week

MATERIAL TESTING

|Type B process



MATERIAL DEVELOPEMENT

Bio Material update

Casting Shape

The wasted coffee mixture can be cast into both thin and thick shapes.

Thin shapes tend to be stiff and brittle, making them prone to breaking.

Introducing a flowing texture can enhance the strength of thin castings, as it allows the material to absorb more force—useful for furniture that must withstand high loads.

Drying Duration and Shape

Longer drying times result in shrinkage, causing the pieces to become smaller in size.

Extended drying can also lead to changes in form, affecting dimensional stability.

Accelerating the Drying Process and Strengthening Properties

To reduce drying time and improve strength:

Decrease the water content by 30%.
Increase protein content in the mixture.

Failed Translucent Coffee Experiment

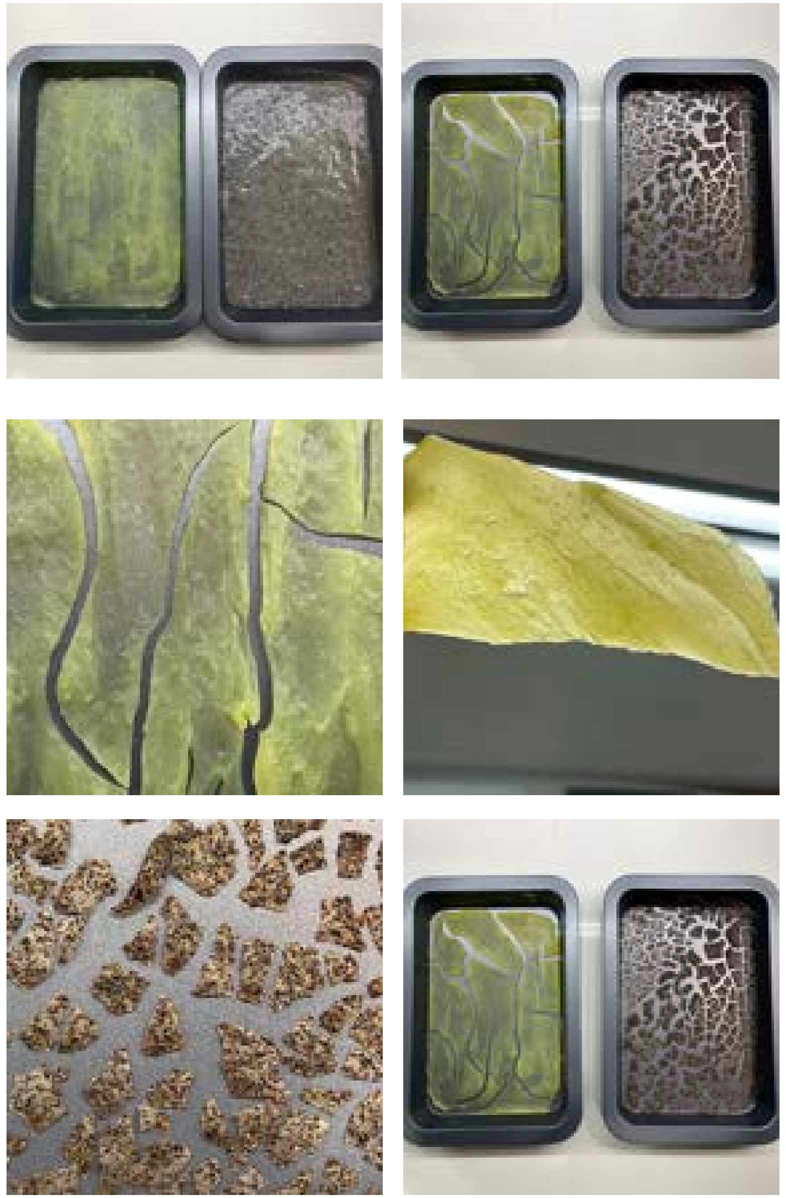
Coffee grounds disrupted the bonding of the bioplastic film, preventing the desired translucency

Week 01 .... **Week 03**

TYPE A : SOLID APPORACH

Week 01 .... **Week 03**

TYPE B : TRANSLUCENT APPORACH



DIGITAL GEOMETRY STUDY

ADDITIONAL REFERENCES

Source :https://www.stevenhaulenbeek.com/icecastbronze

COMPLEX MODULAR

DIGITAL GEOMETRY STUDY / INITIAL DESIGN IDEAS

The example of posible geometric systems, regarding to fabrication process that showcase the possibilities of casting.

AR/MR METHOD

AR/MR study : Human in Loop

The core idea is a "human-in-the-loop" system, where human knowledge and aesthetic judgment are integrated with the precision of robotic fabrication. This setup, referred to as a Human-Cyber-Physical System
 (HCPS), combines three main components:

1. Form Generation:

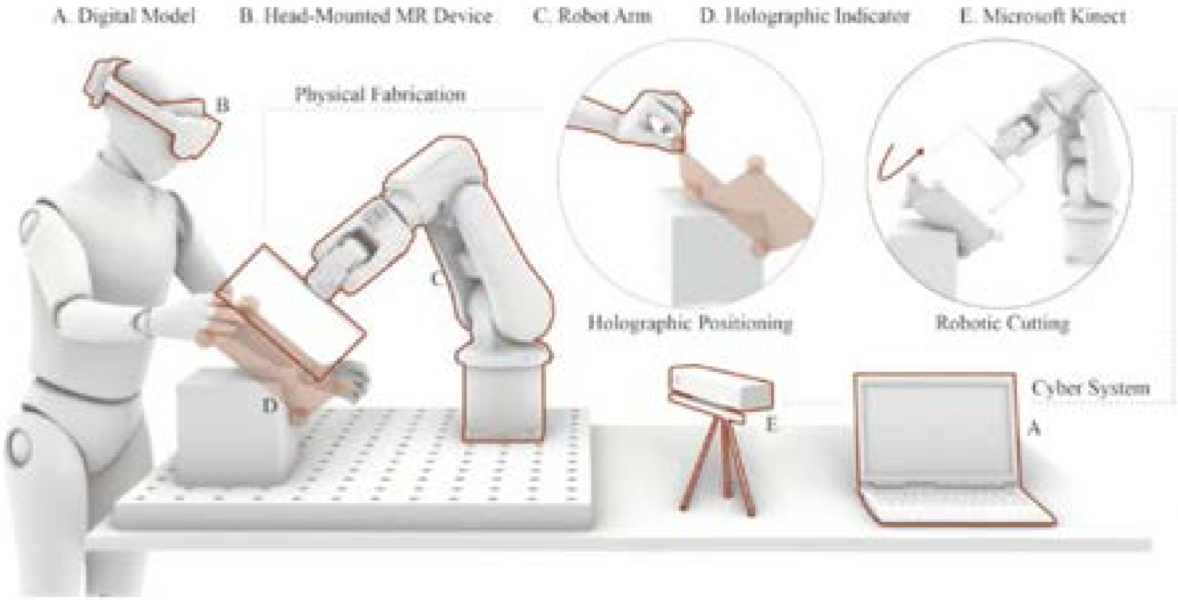
 A natural Taihu rock was 3D-scanned to analyze its geometric features ("slender, wrinkle, clear, leaky"). These features were translated into a shape grammar—a set of rules used to computationally generate a design for a new rock form.

2. Robotic Fabrication:

The design was then used to create robotic foam-cutting paths. The fabrication system involved an industrial robot with various cutting tools.

3. Mixed Reality (MR) Augmentation:

Using a Head-Mounted Display (HMD), a human operator could see a holographic overlay of the digital model on the physical EPS block in real-time. This allowed them to make intuitive, on-the-spot adjustments to the robot's cutting path based on their aesthetic judgment. This bypasses the need for constant mouse and keyboard input, creating a more natural and fluid interaction.



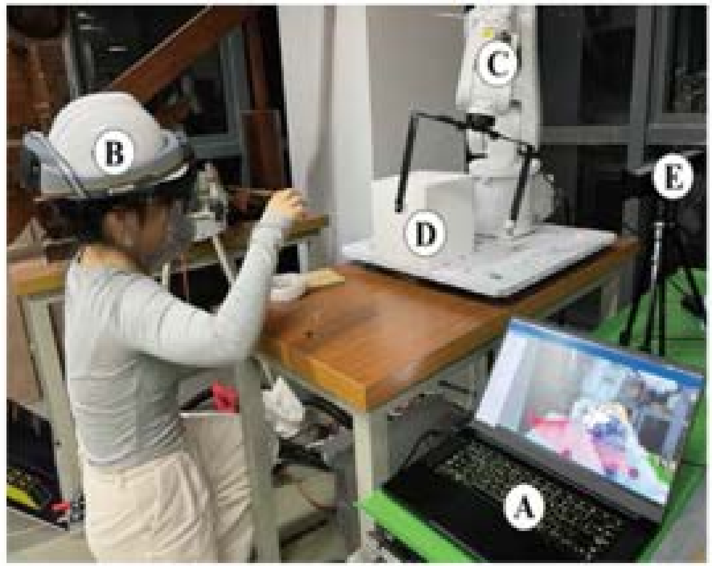


Fig. 3. 3D scan of a natural Taihu rock and the analysis of its geometric characteristics.

https://doi.org/10.1007/978-981-99-8405-3_29

AR/MR METHOD

How AR/MR integrated with the project

1.Digital Design: A 3D model of the final object is created.

This model incorporates shape grammar rules based on a desired aesthetic, and it accounts for material properties such as shrinkage.

2.MR-Aided Mold Preparation:

Using a Head-Mounted Display (HMD), a holographic version of the mold design is projected directly onto the workspace. This provides a live, 3D blueprint that guides the human fabricator.

3.Human-Guided Fabrication:

The human sculpts the mold-making material (e.g., silicone or plaster) according to the holographic guide. The AR system can track gestures, allowing the human to intuitively alter the virtual model in real-time. This dynamic feedback loop combines the precision of the digital model with the unique, expressive qualities of human craftsmanship.

4.Verification and Casting:

The MR interface can be used to perform a final check of the mold, identifying potential flaws before pouring the wasted coffee ground composite.

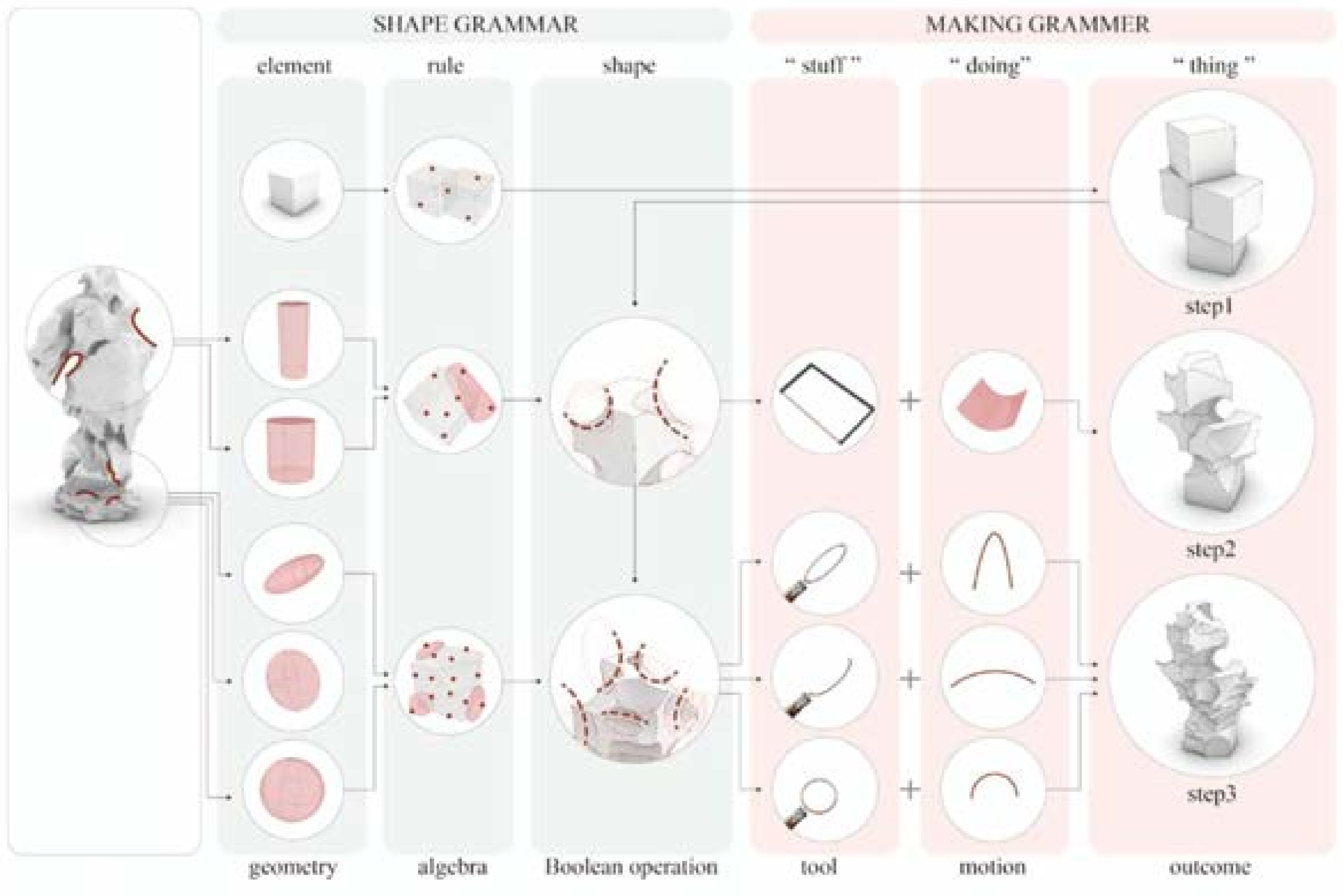


Fig. 2. Design and materialization setup of a shape grammar formed Taihu rock.

https://doi.org/10.1007/978-981-99-8405-3_29

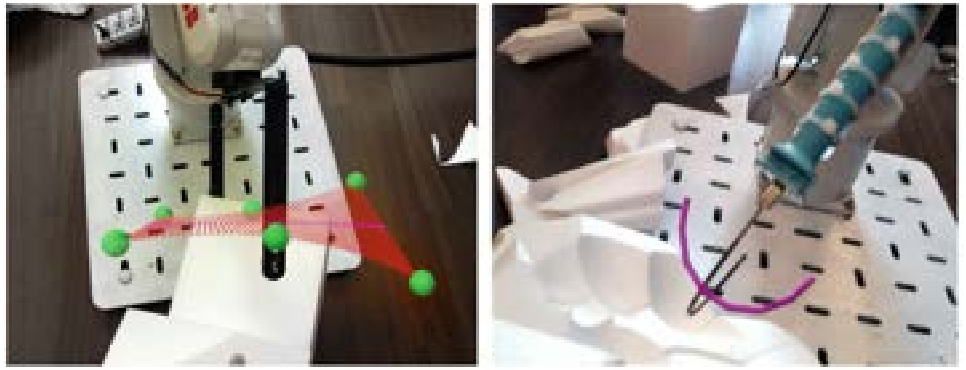


Fig. 4. Human-guided robotic fabrication with customized tools. Rough cut with hyperbolic paraboloid (left) and fine carving with trajectories (right).

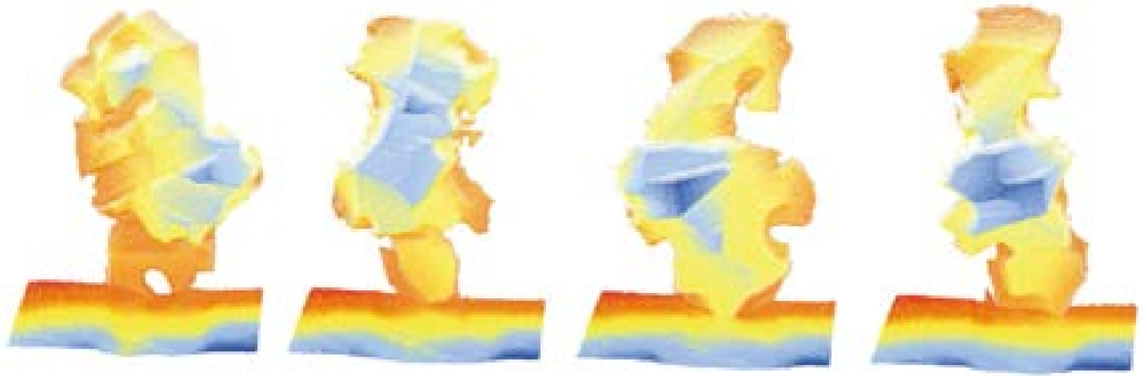


Fig. 6. In-progress scan using Kinect to evaluate geometric complexity.

Fig. 5. MR-aided robotic fabrication allows for a safe human-machine collaboration envi-ronment and flexible system setup.

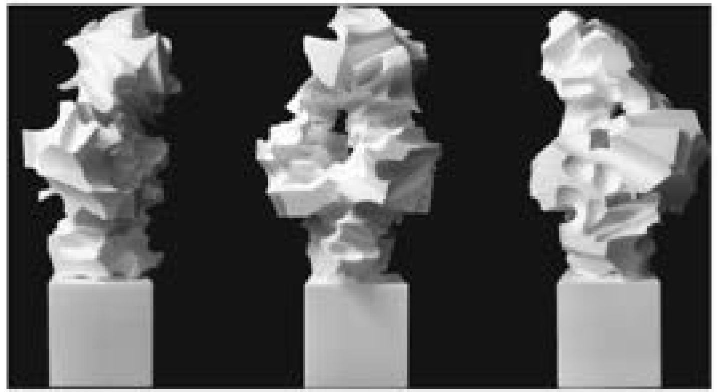
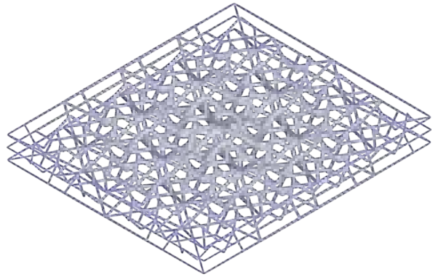
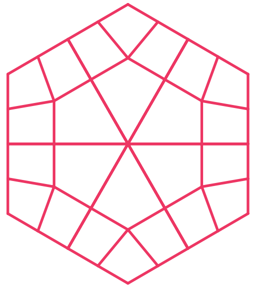


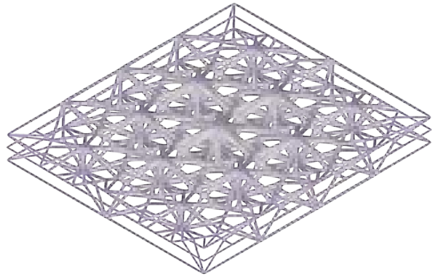
Fig. 7. The built outcome of EPS-made Taihu Rock.

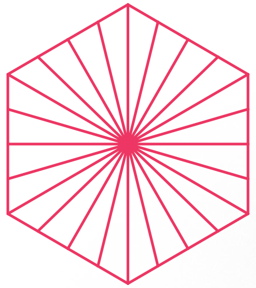
https://doi.org/10.1007/978-981-99-8405-3_29

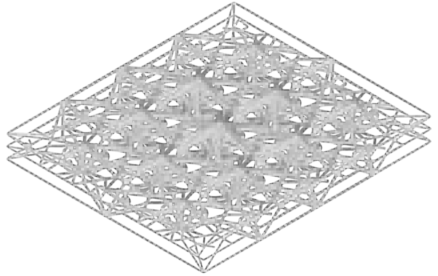
DIGITAL GEOMETRIC DESIGN
Lattice strcture

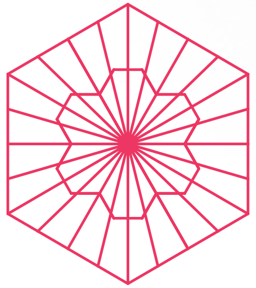


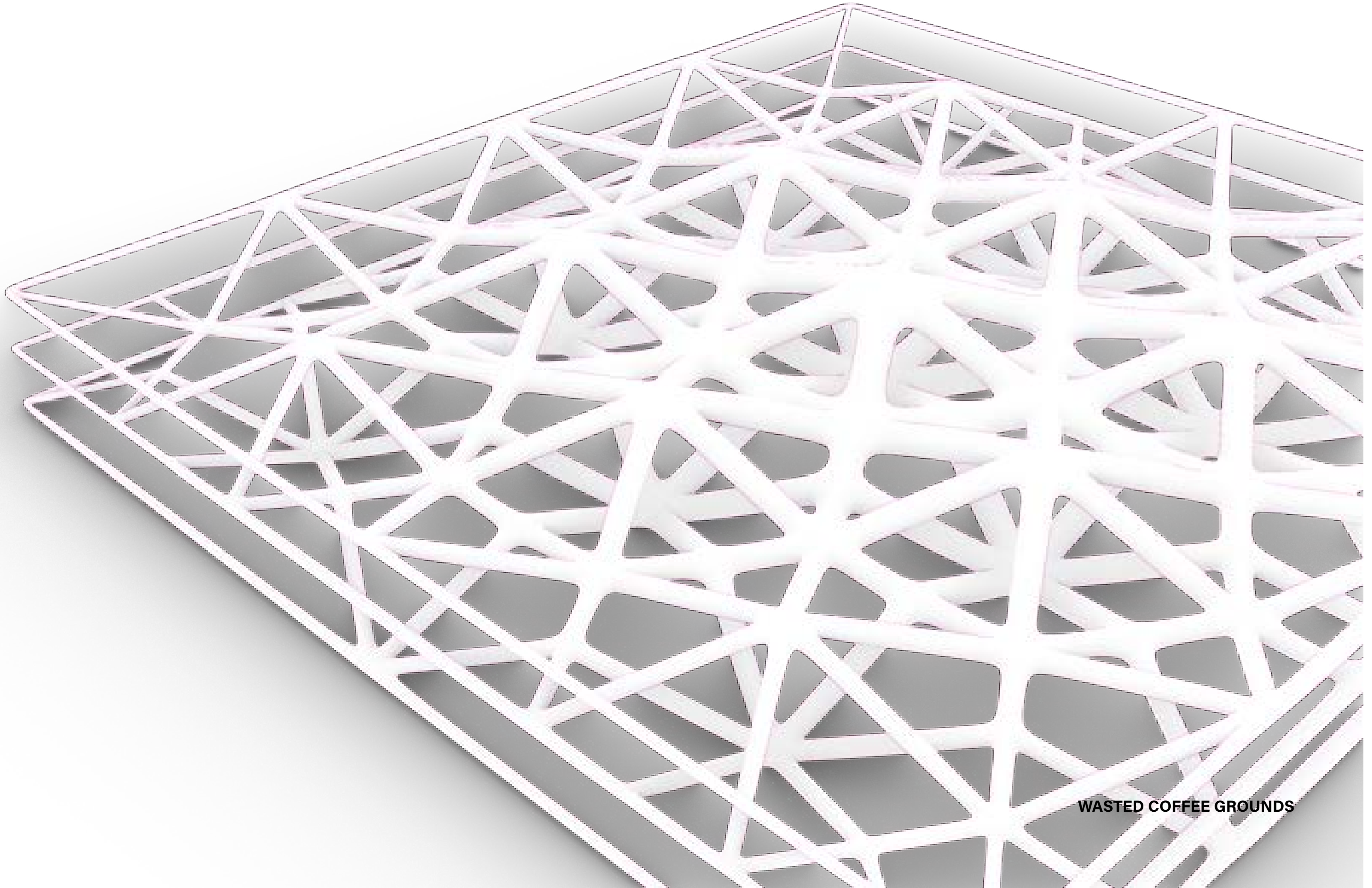






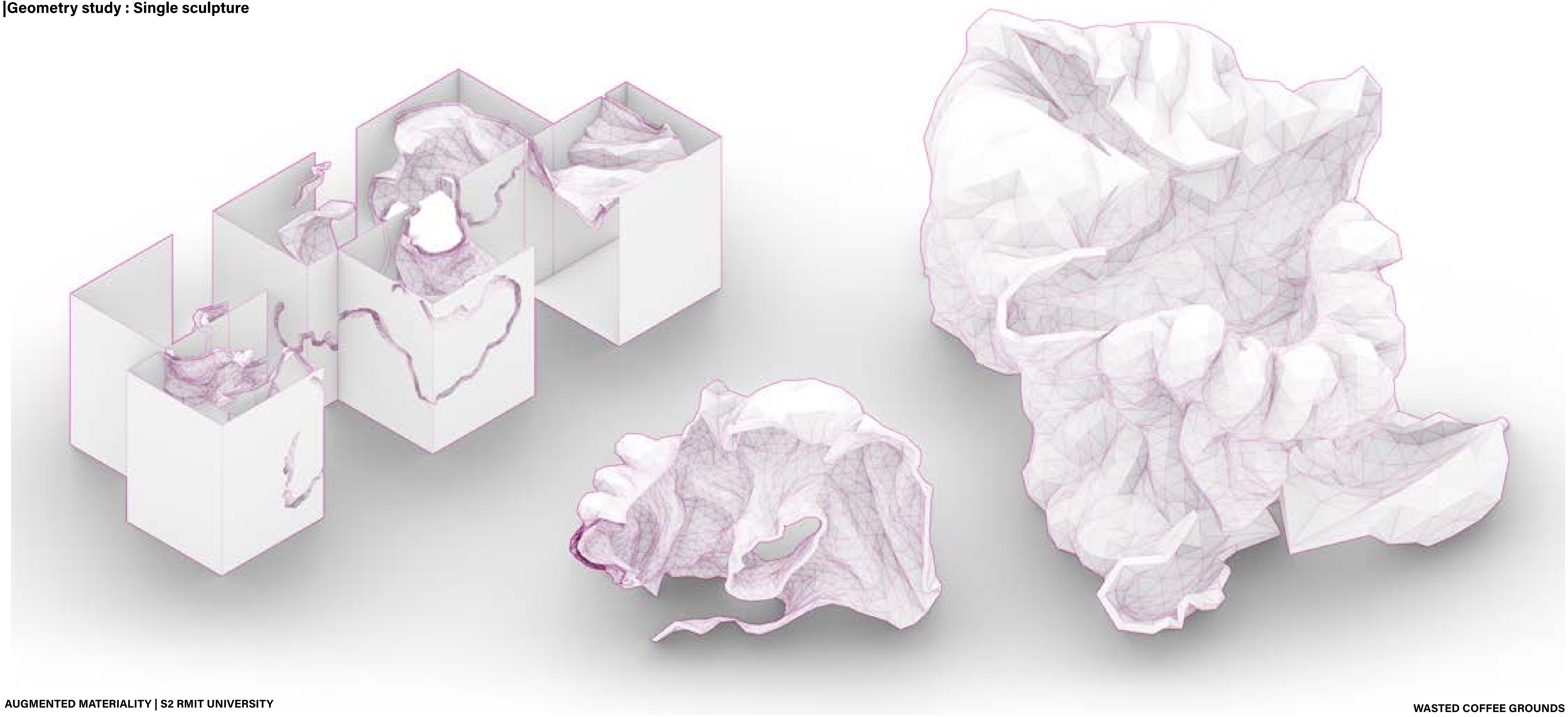






DIGITAL GEOMETRIC DESIGN

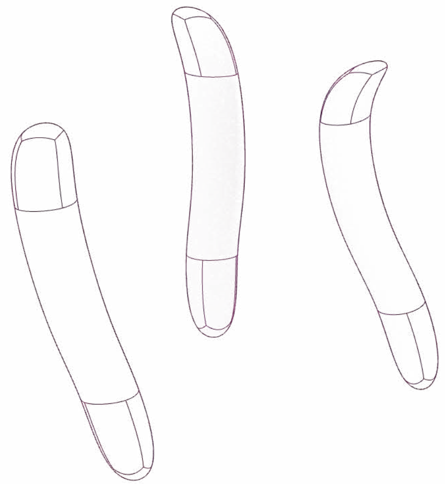
|Geometry study : Single sculpture

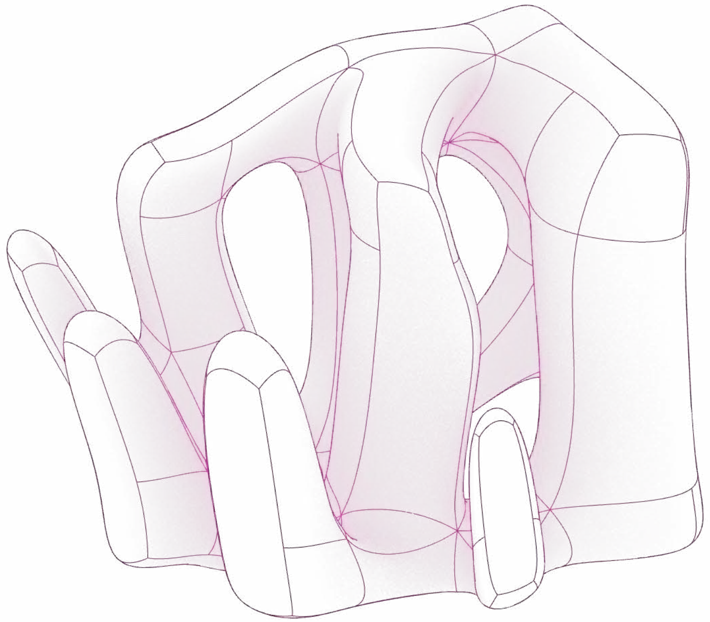


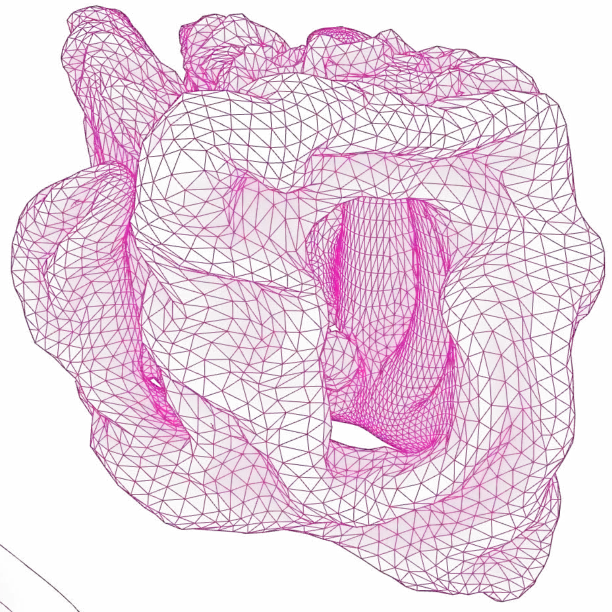
DIGITAL GEOMETRIC DESIGN

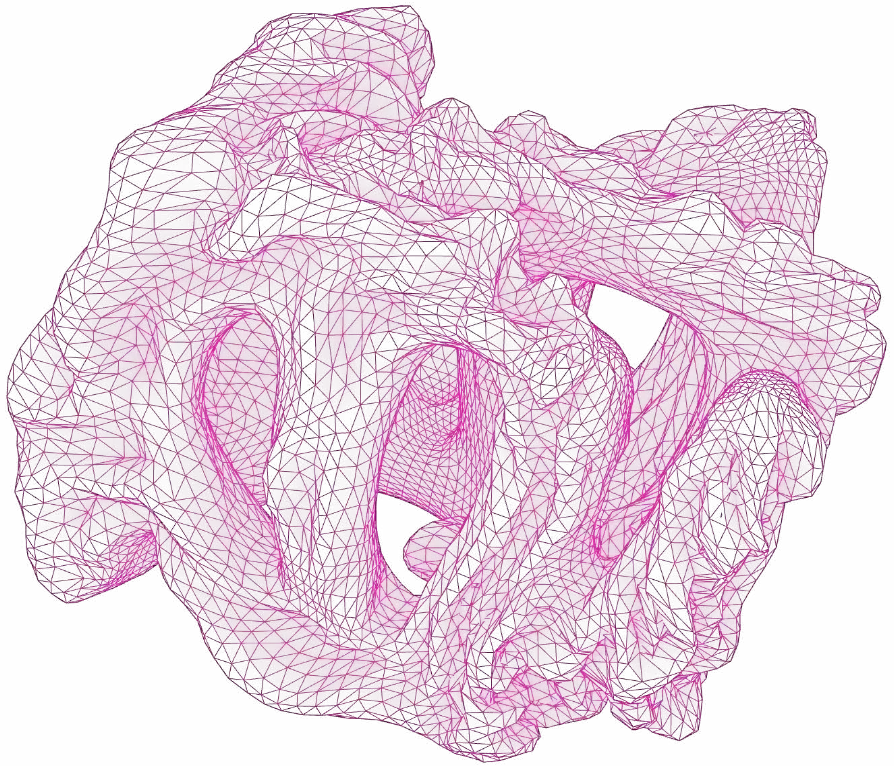
Geometry study : Coral

//BASE GEOMETRY





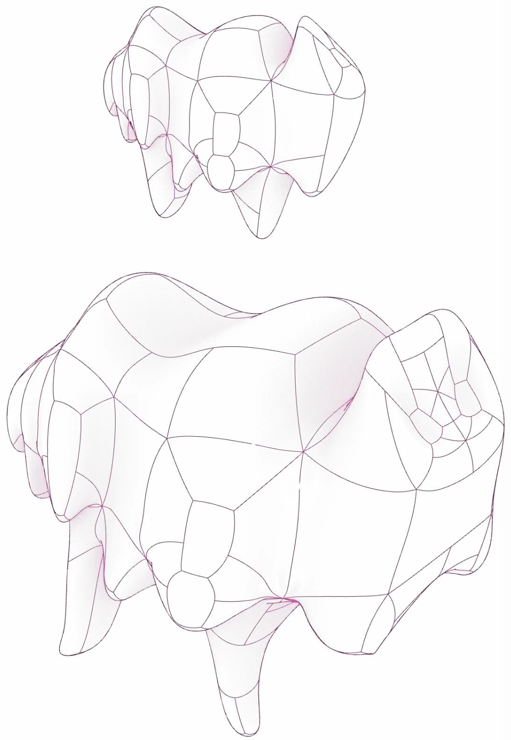


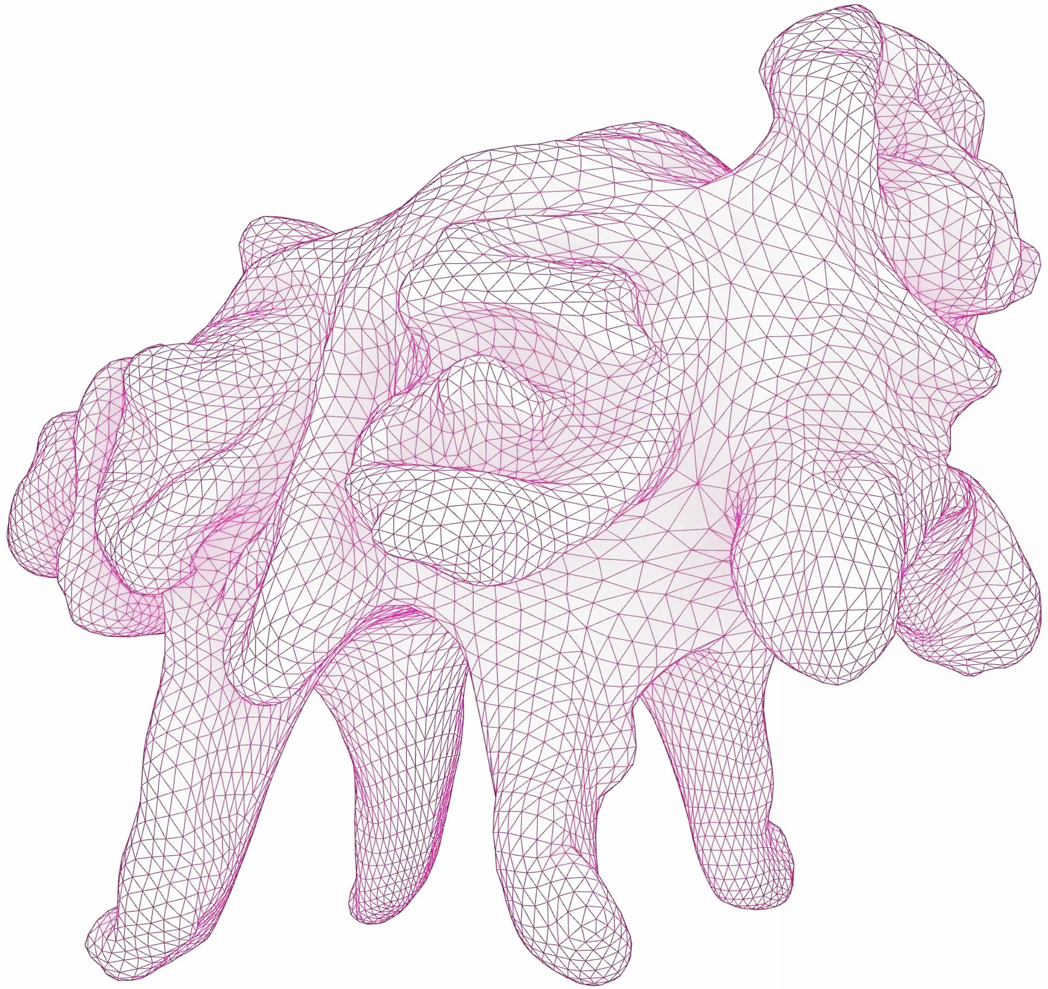


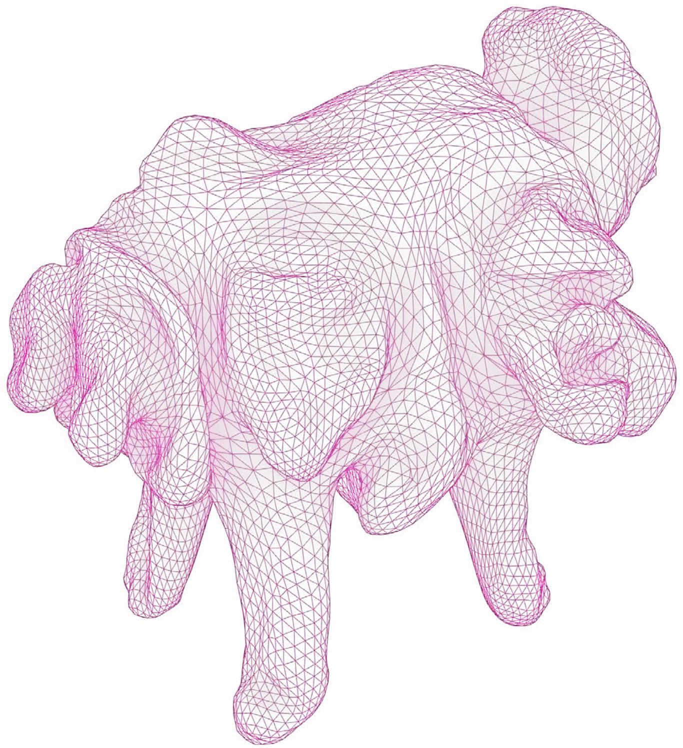
DIGITAL GEOMETRIC DESIGN

Geometry study : Abstract 1

//BASE GEOMETRY

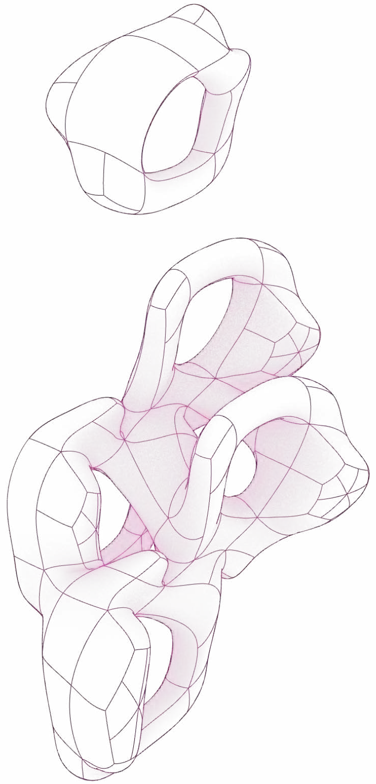


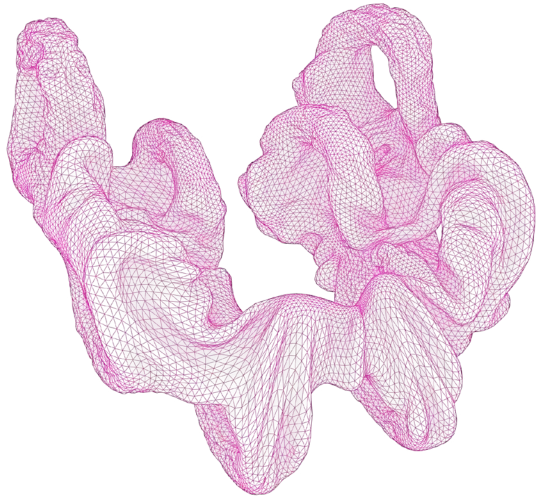




DIGITAL GEOMETRIC DESIGN
Geometry study : Abstract 2

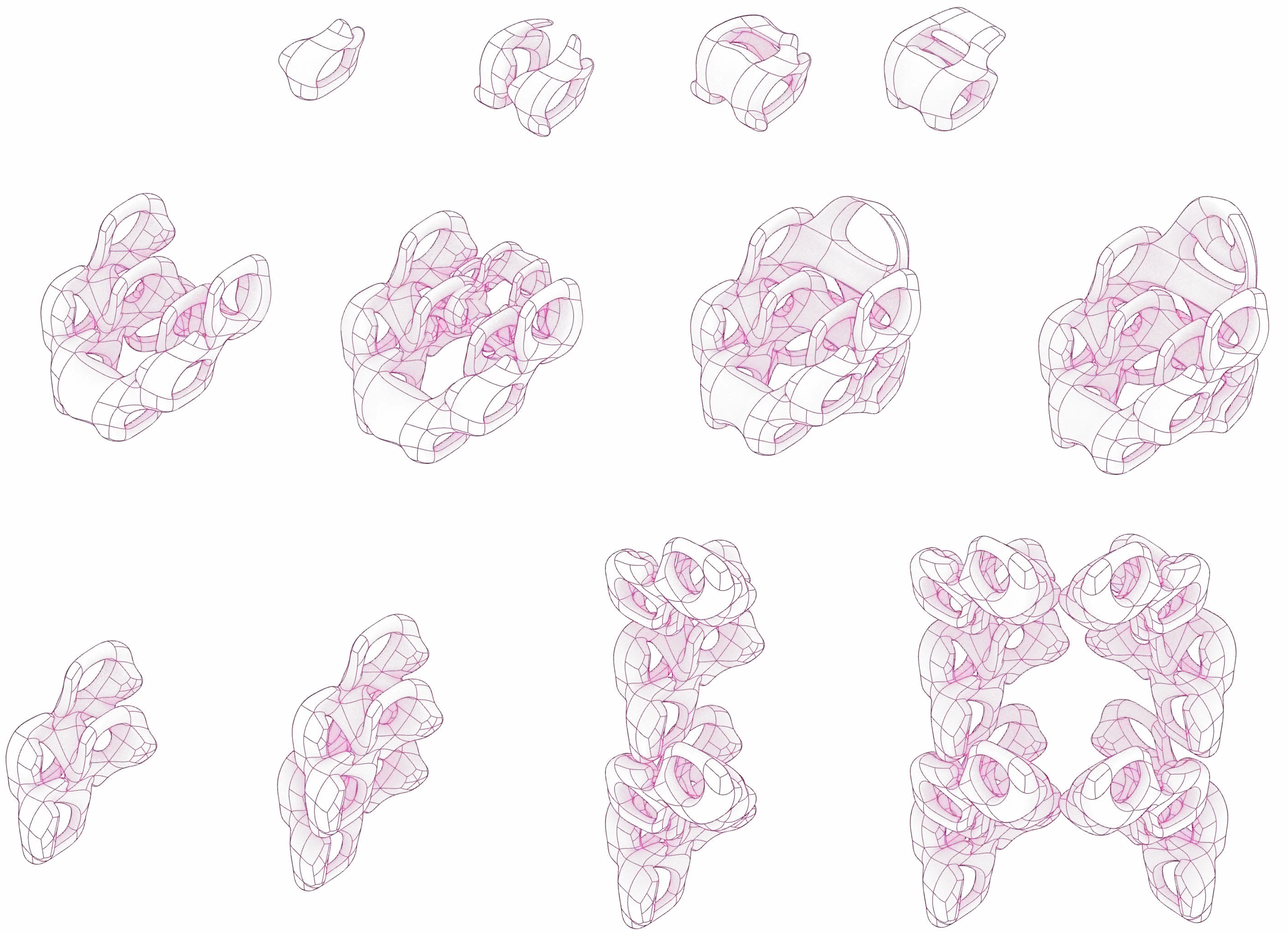
//BASE GEOMETRY



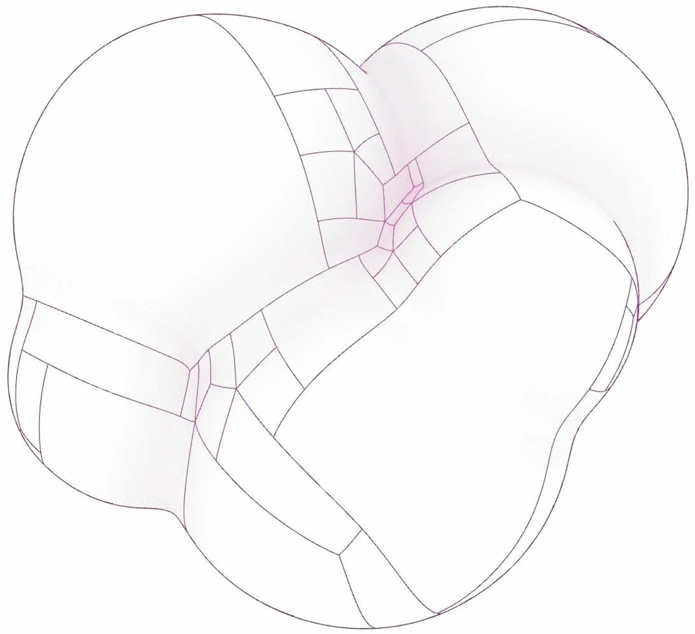


DIGITAL GEOMETRIC DESIGN

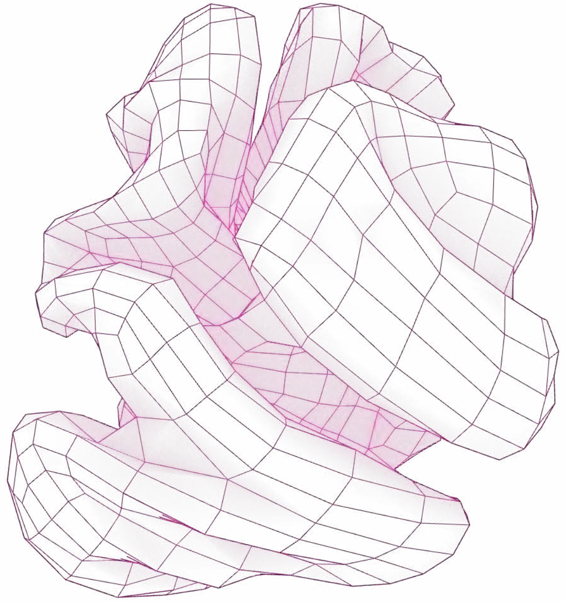
Geometry study : Schematic design



SIMPLIFY GEOMETRY



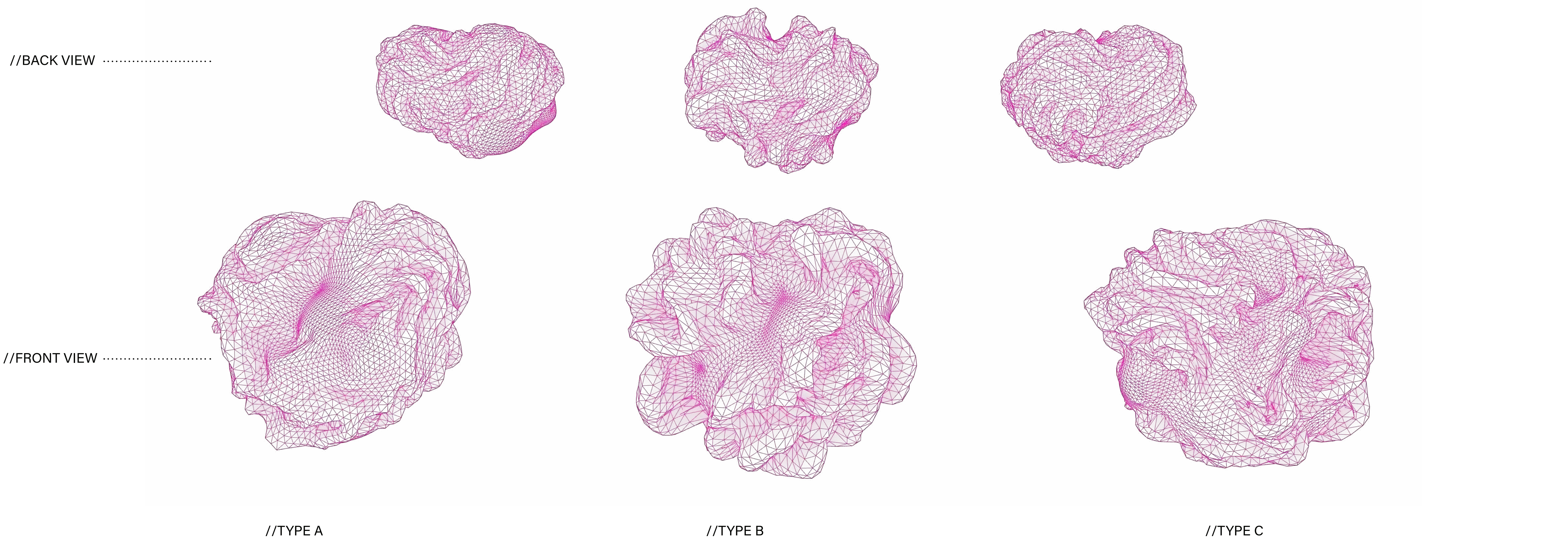
//BASE GEOMETRY



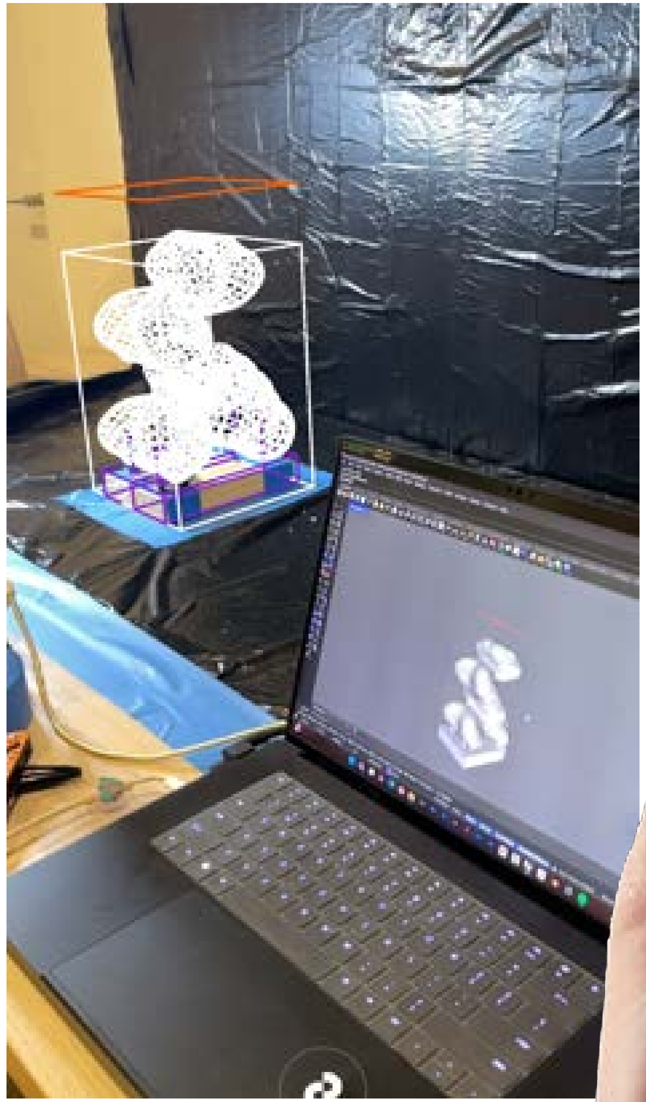
//SIMPLE GEOMETRY

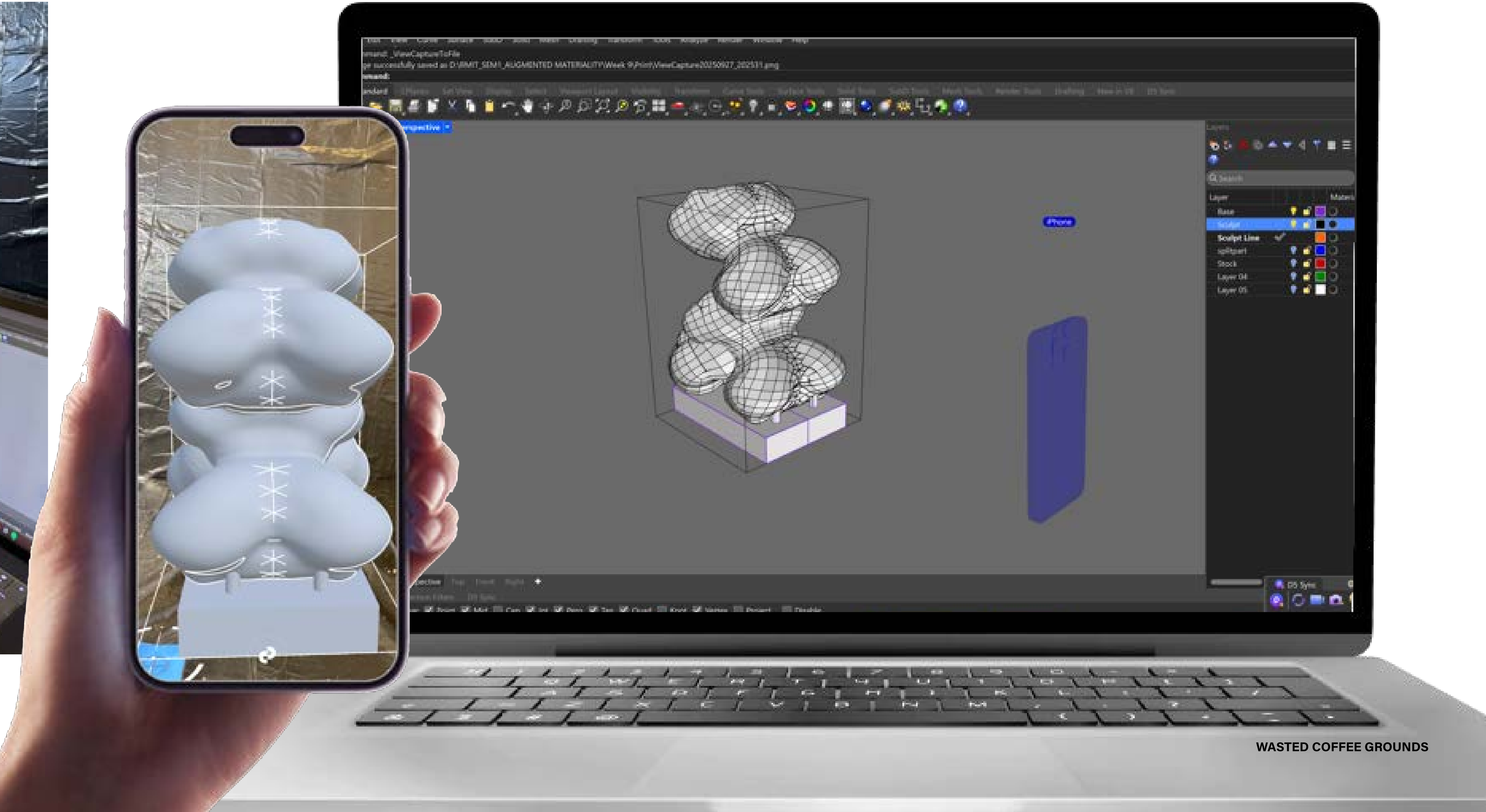
DIGITAL GEOMETRIC DESIGN

SIMPLIFY GEOMETRY
(High Resolution for detailed crafting)

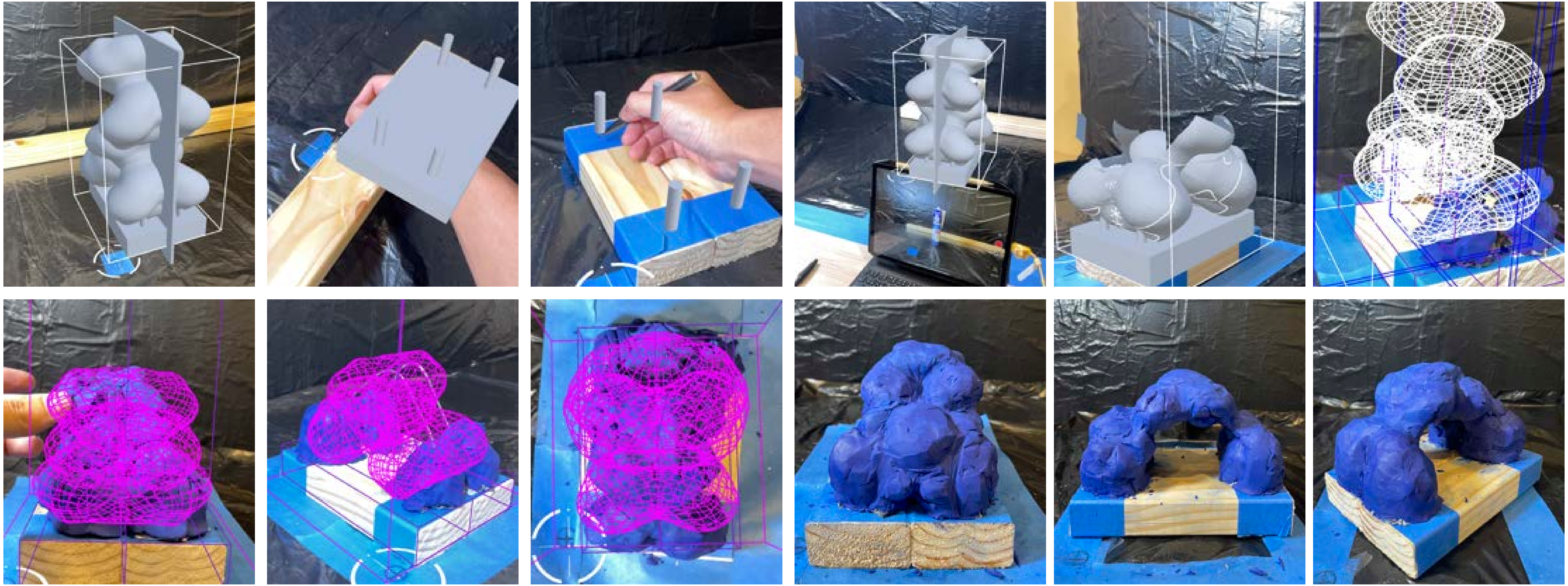


AR/MR WORKFLOW
Using fologram for base geometry guideline





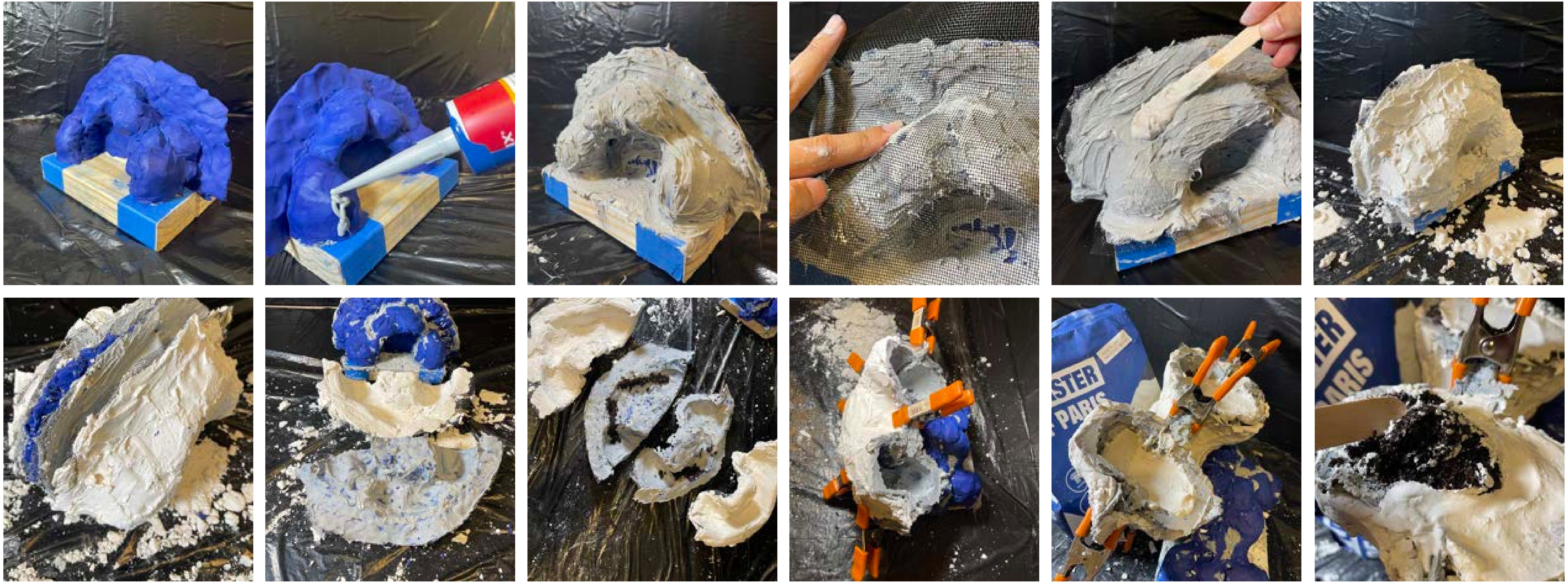
AR/MR WORKFLOW
Using fologram for base geometry guideline



CASTING TEST

Positive moulding

Base geometry (oil clay) --> Silicone mould (2 days curing) --> Plaster mould (2 hours) -->Material casting (1 day)



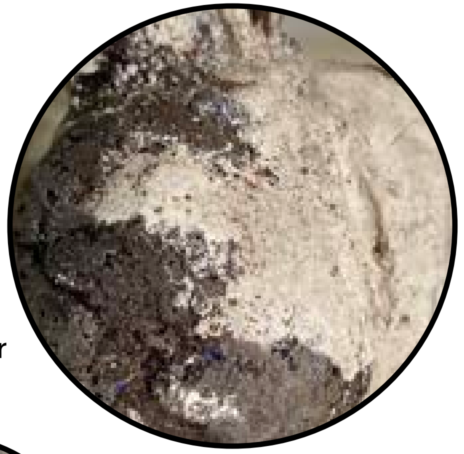
CASTING TEST

Composite material casting test :
Solid coffee & polyurethane mixture (AH) and Plaster coffee (CA)

Demolding

CASTING TEST

Material’s characteristic



Material transition between coffee and plaster

Using plaster as main material,
while coffee only add texture.

Raw coffee material gesture

Cracking part shows
inner layer of coffee material.

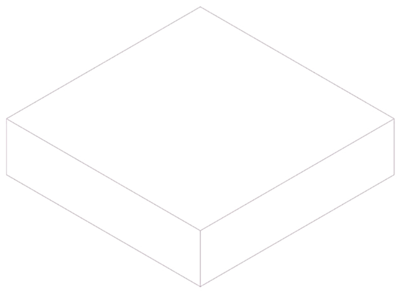
Smooth coffee material gesture
shape edge condition

GEOMETRY DESIGN 1 : WAVY PANEL

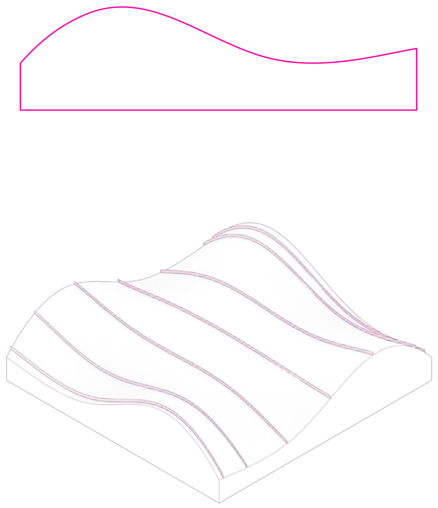
Connect edge panel

The approach was to create 1–2 conect edge panels that can utilize repetition technique to create interesting geometry.

The modeling process was straightforward. It mostly involved simple curves and patch command in rhino, which I found to be repeatable.

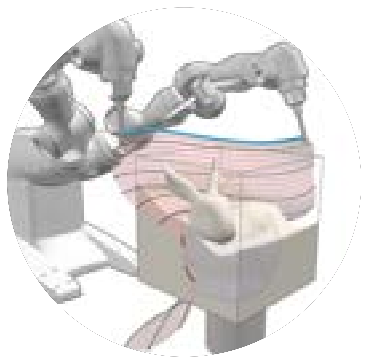


BASE GEOMETRY

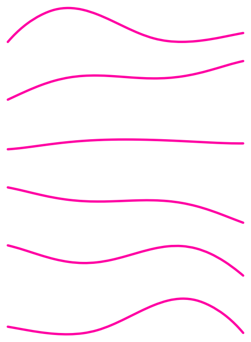


BOOLEAN OPERATION (CUTTING PROFILE)

initial phase



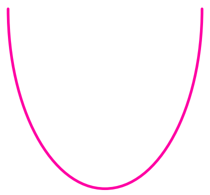
blending hotwire



Using layout guildline AR/MR

Detail phase

grooving hotwire



Overlay detail typology guild-line by usingAR/MR

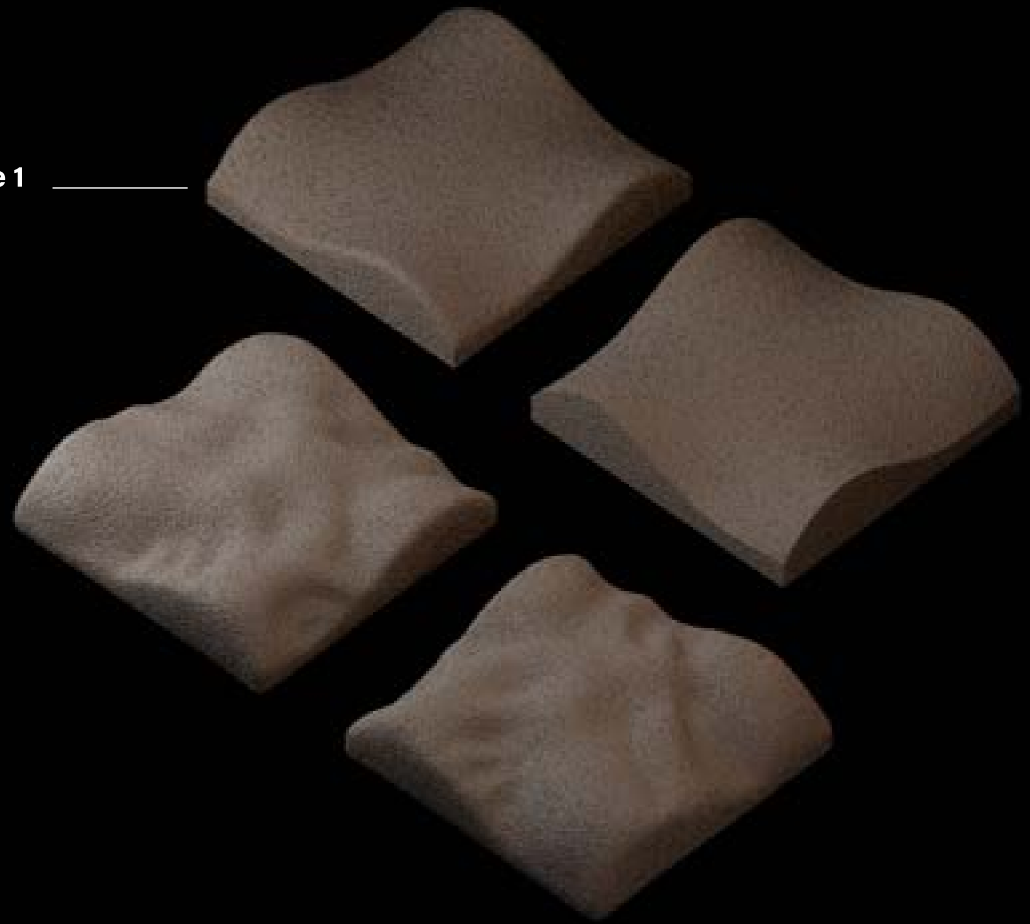
TOOLS

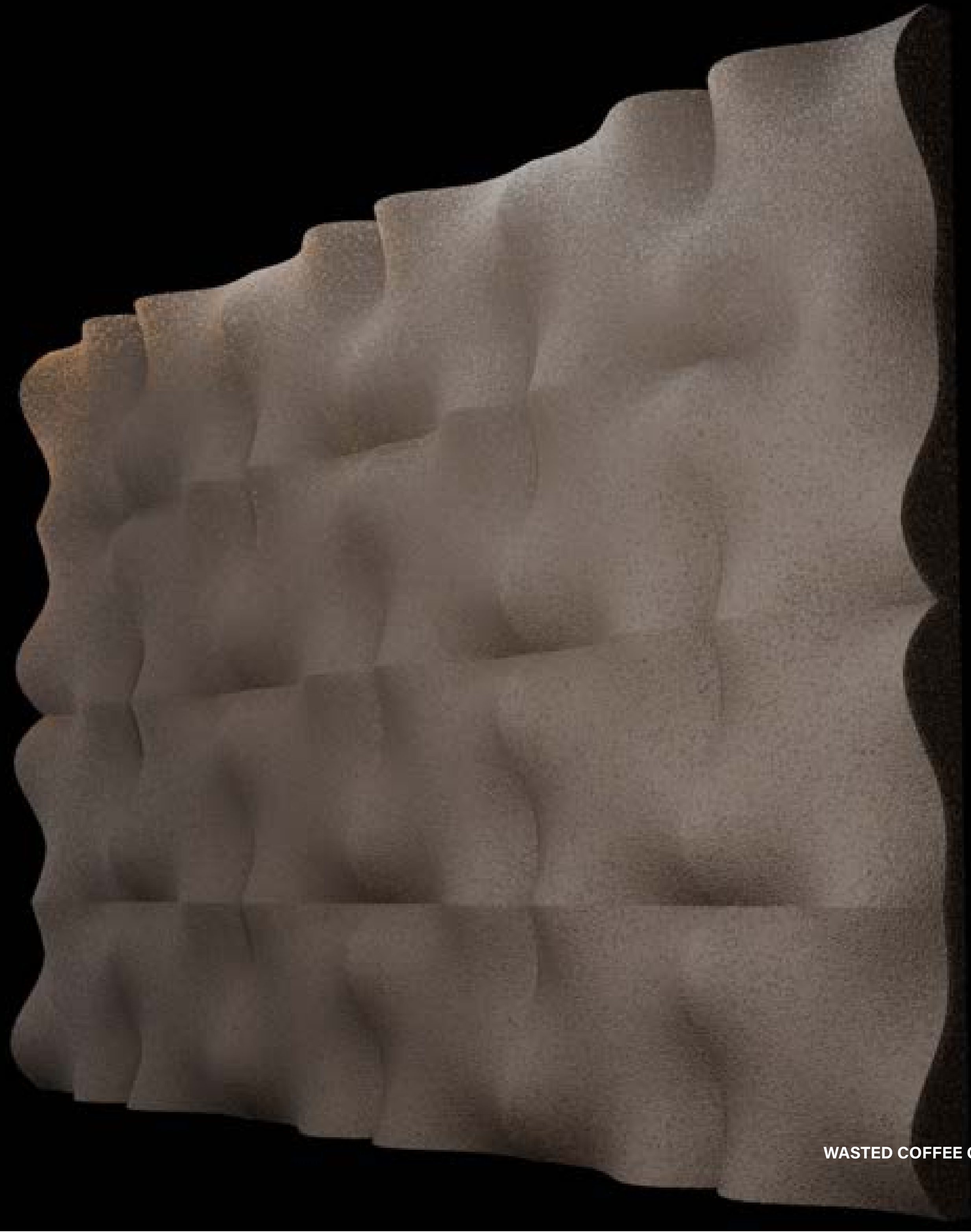
MOTION

RESULTS

GEOMETRY DESIGN 1 : WAVY

Single coffee material





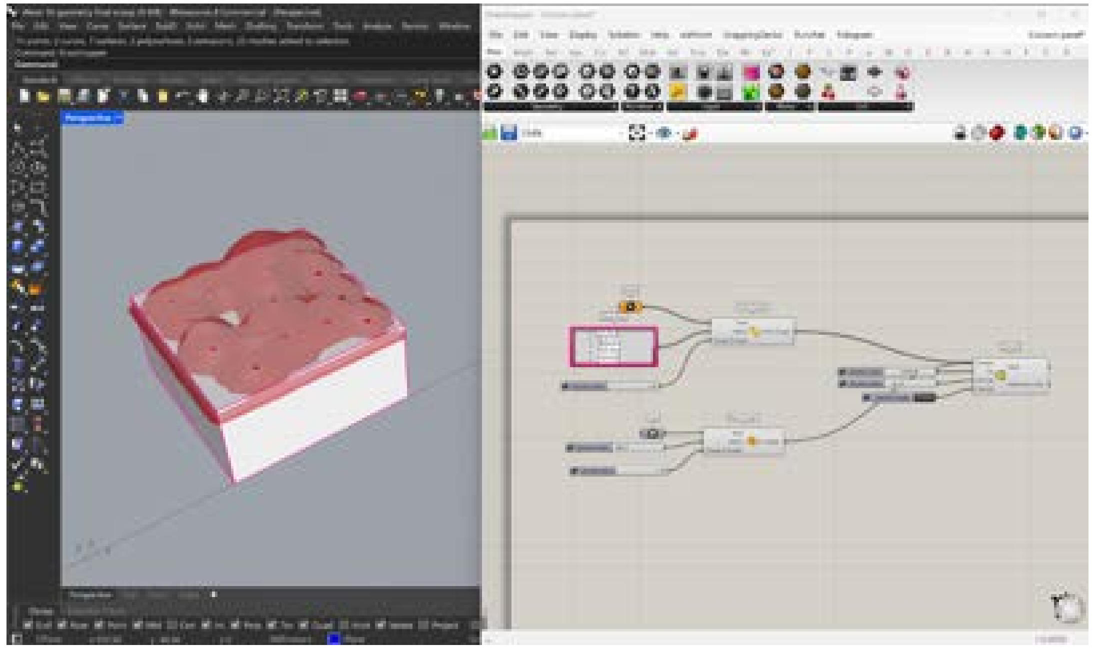
GEOMETRY DESIGN 1 : WAVY

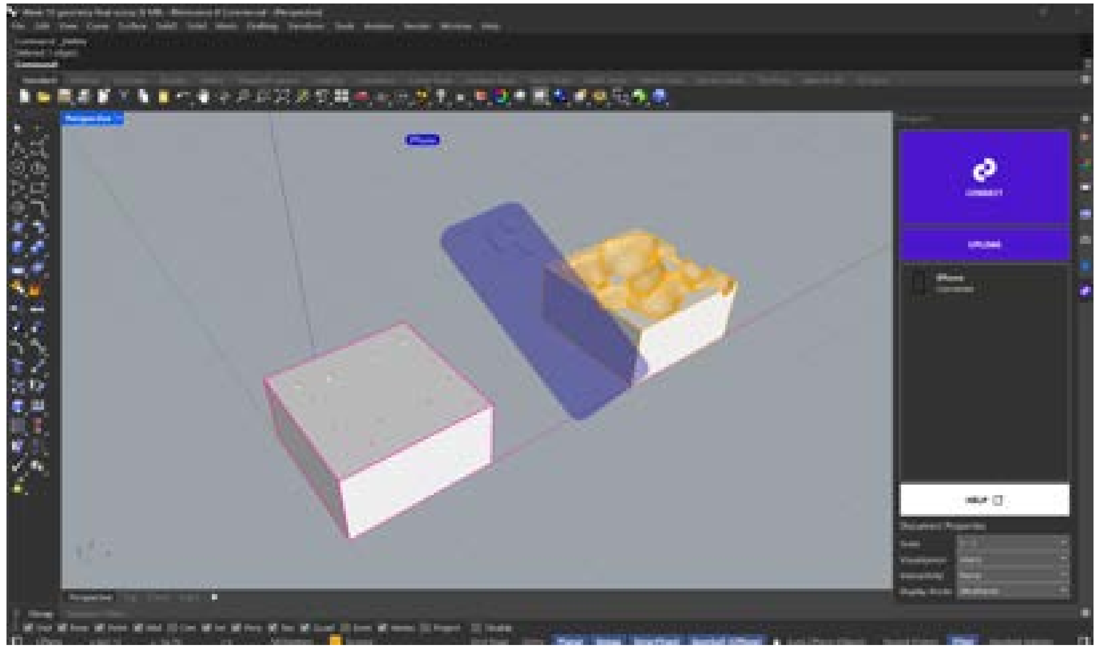
Single coffee material

GEOMETRY DESIGN 2 : BLOBY PILLOW

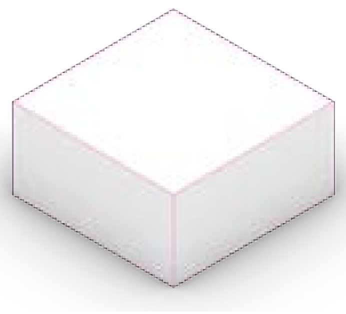
Disconnect edge panel

Disconnected “Blobby Pillow” Panel
The overall form resembled a pillow, which originally generated from a Cocoon in Grasshopper. With hotwire scoop tool could achieve this shape.

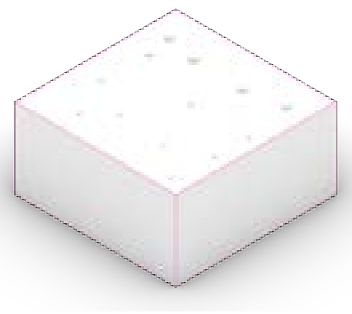




PROCESS



Base geometry



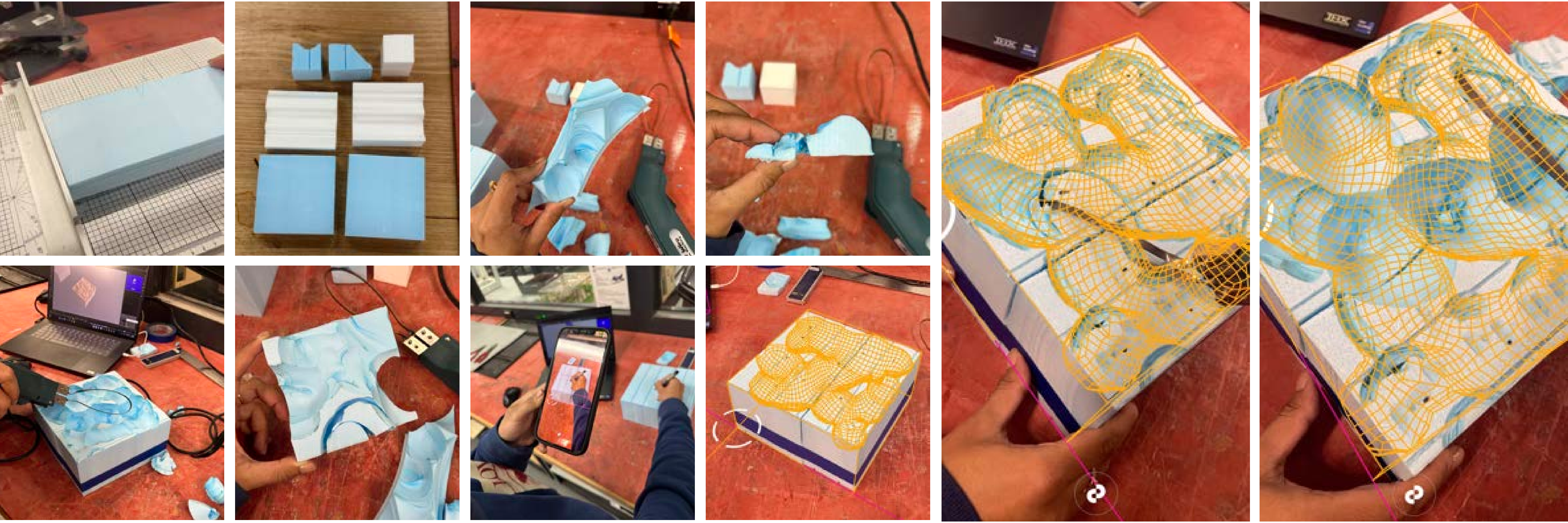
Positioning guideline (AR/MR)

Typology guideline (AR/MR)

Casting

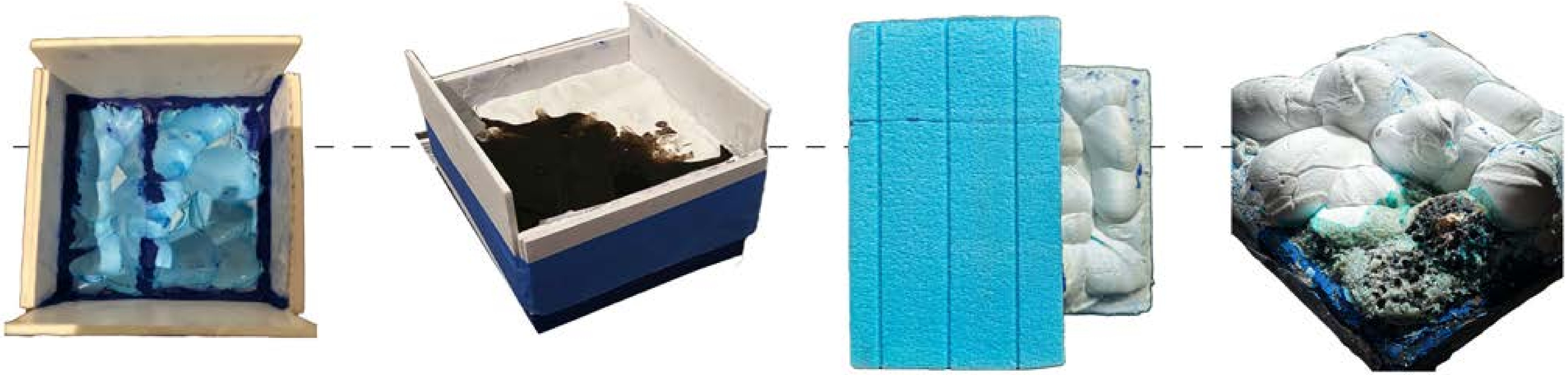
GEOMETRY DESIGN 2 : BLOBBY PILLOW

Disconnect edge panel

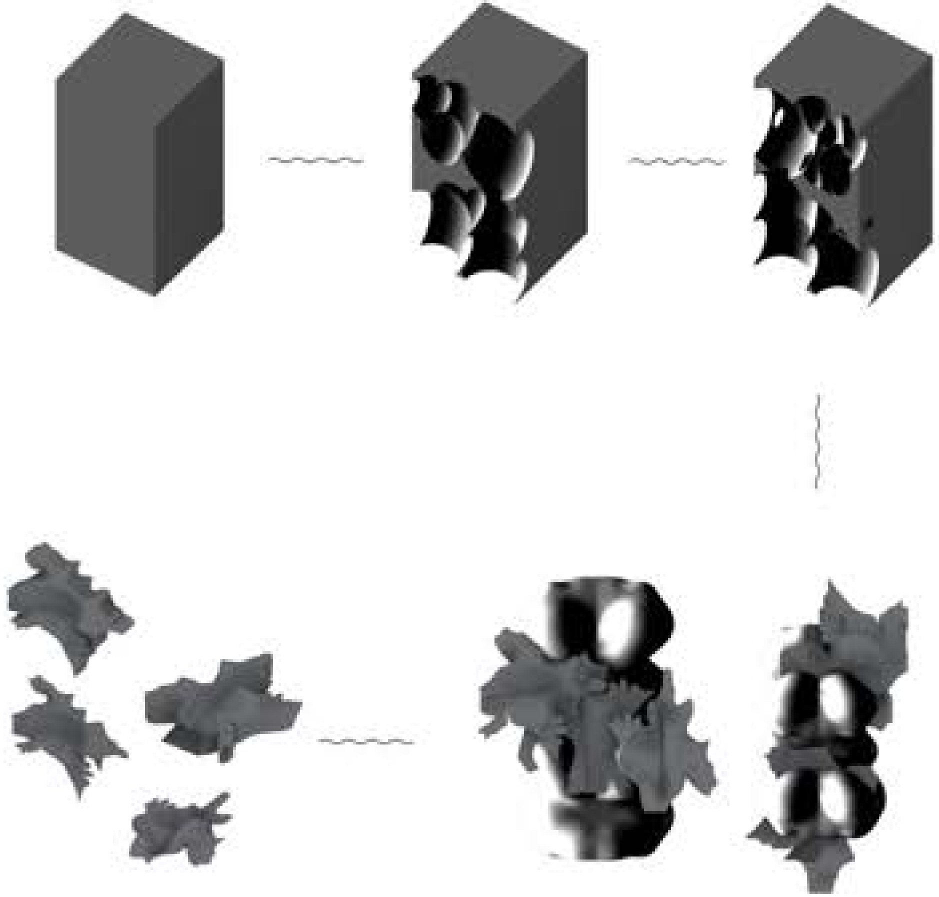


CASTING TEST

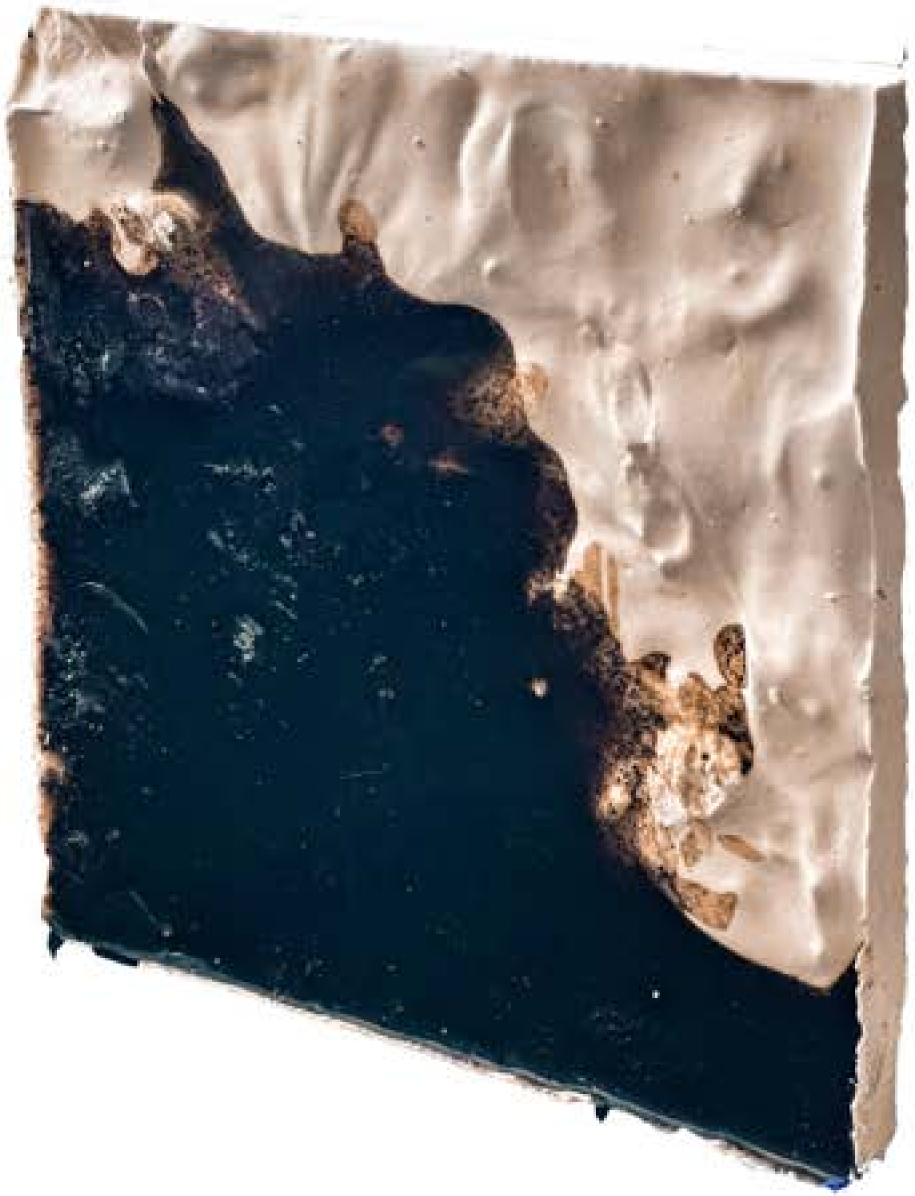
The process



NEGATIVE CASTING :



FAILED ATTEMPT



TEXTURES CLOSEUP

GEOMETRY DESIGN 2 : BLOBY PILLOW

Single coffee material

GEOMETRY DESIGN 2 : BLOBBY PILLOW

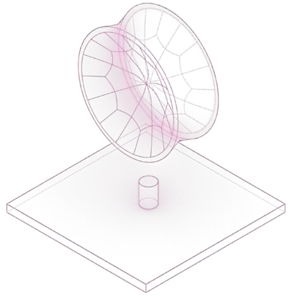
GEOMETRY DESIGN 2 : BLOBBY PILLOW

Composite coffee material

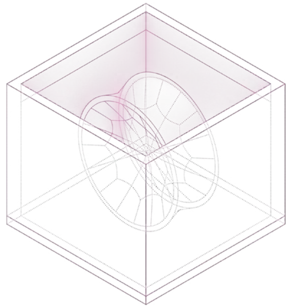
MOULDING & CASTING TEST

Positve geometry
1-part silicone moulding test
2-parts paster moulding test

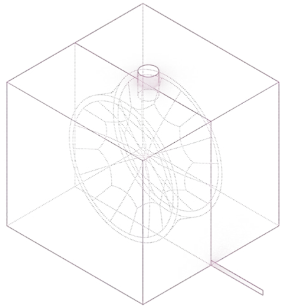
1-PART SILICONE MOULD PROCESS



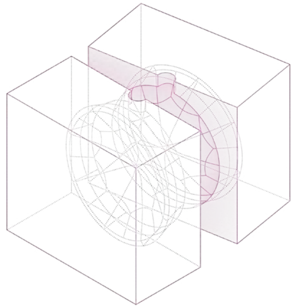
Base geometry setup



Silicone pouring

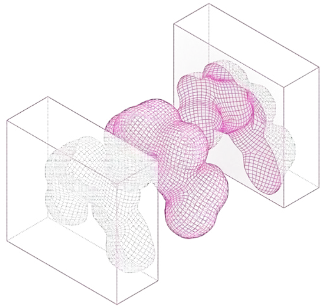


Cutting

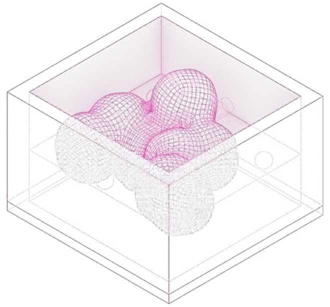


Casting& Demoulding

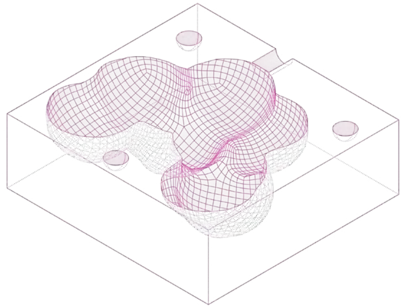
2-PART PLASTER MOULD PROCESS



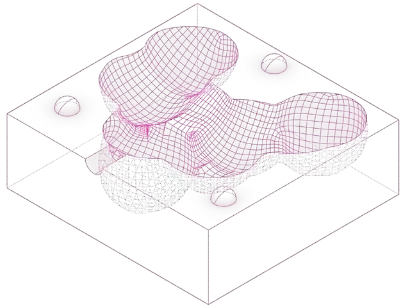
Base geometry



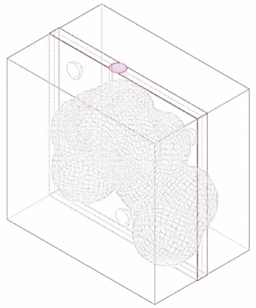
Plaster pouring



Key making

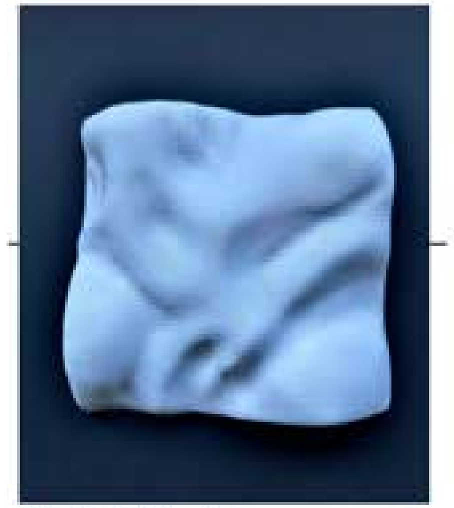


Opposite mould



Casting&Demoulding

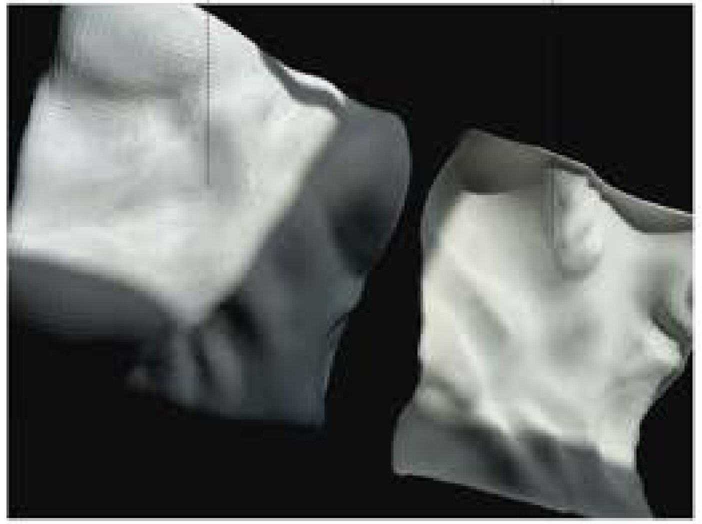
GEOMETRY DESIGN : BLOBBY PILLOW (CASTING ON 3D PRINT)



3d Printed Blobby Pillow

Casted on 3d Print

Precision



Groove Details

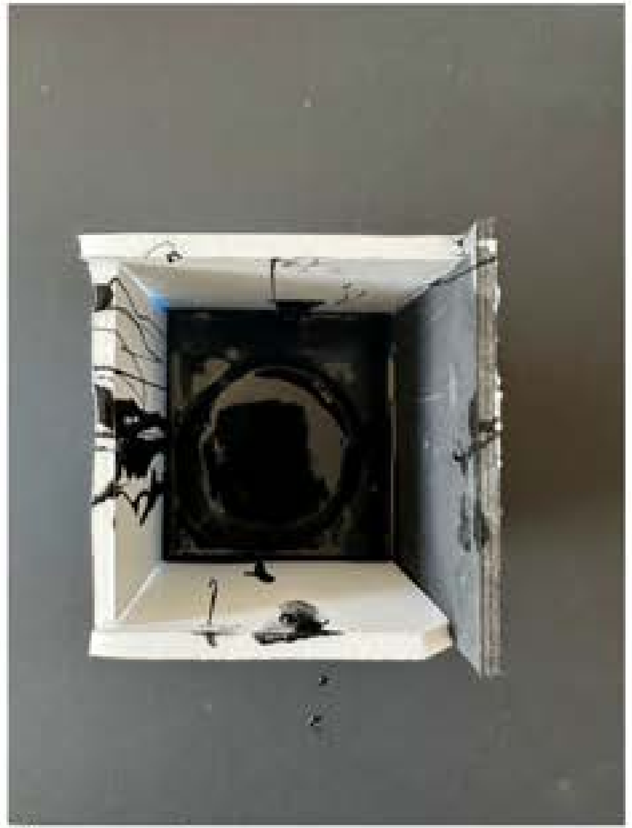
CASTING TEST

Positive molding

Base geometry (oil clay) --> Silicone mould (2 days curing) --> Plaster mould (2 hours) -->Material casting (1 day)

Materials for silicone moulding

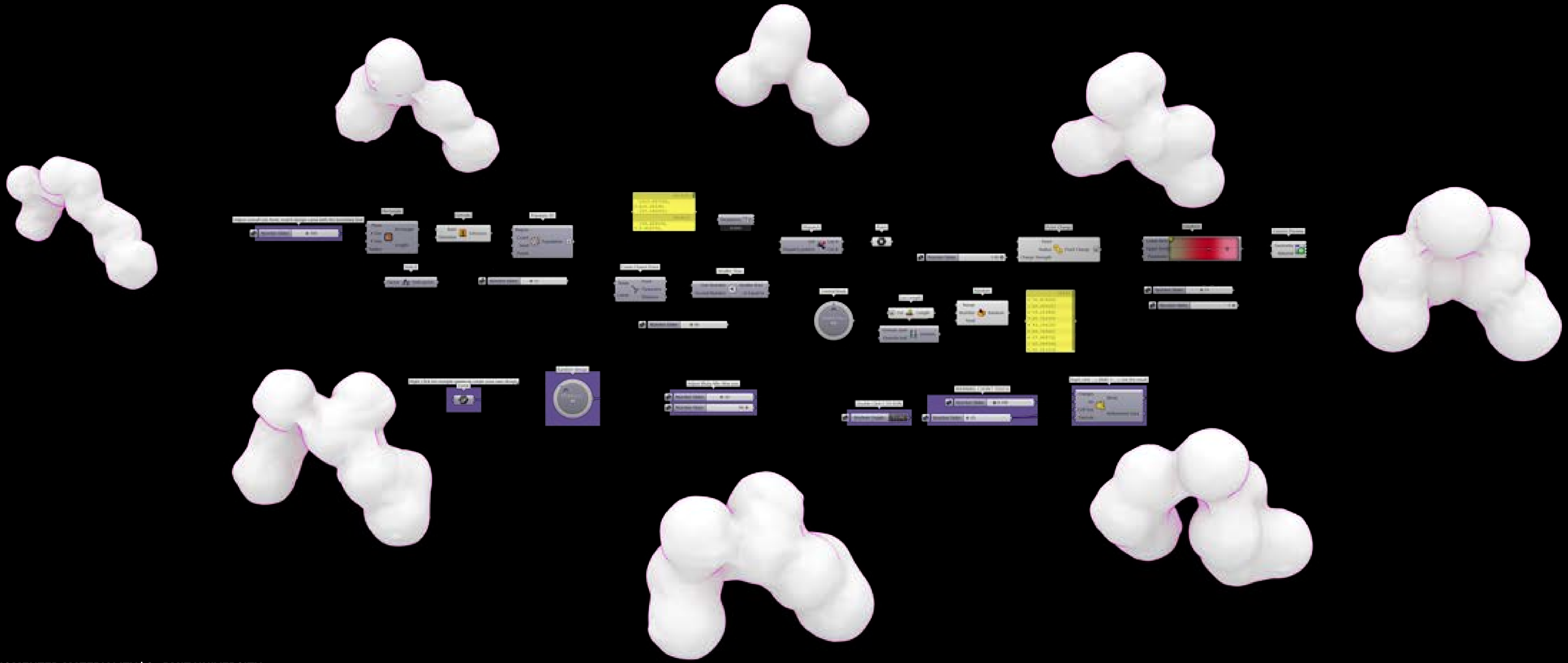
Prepared mixture



Silicone poured

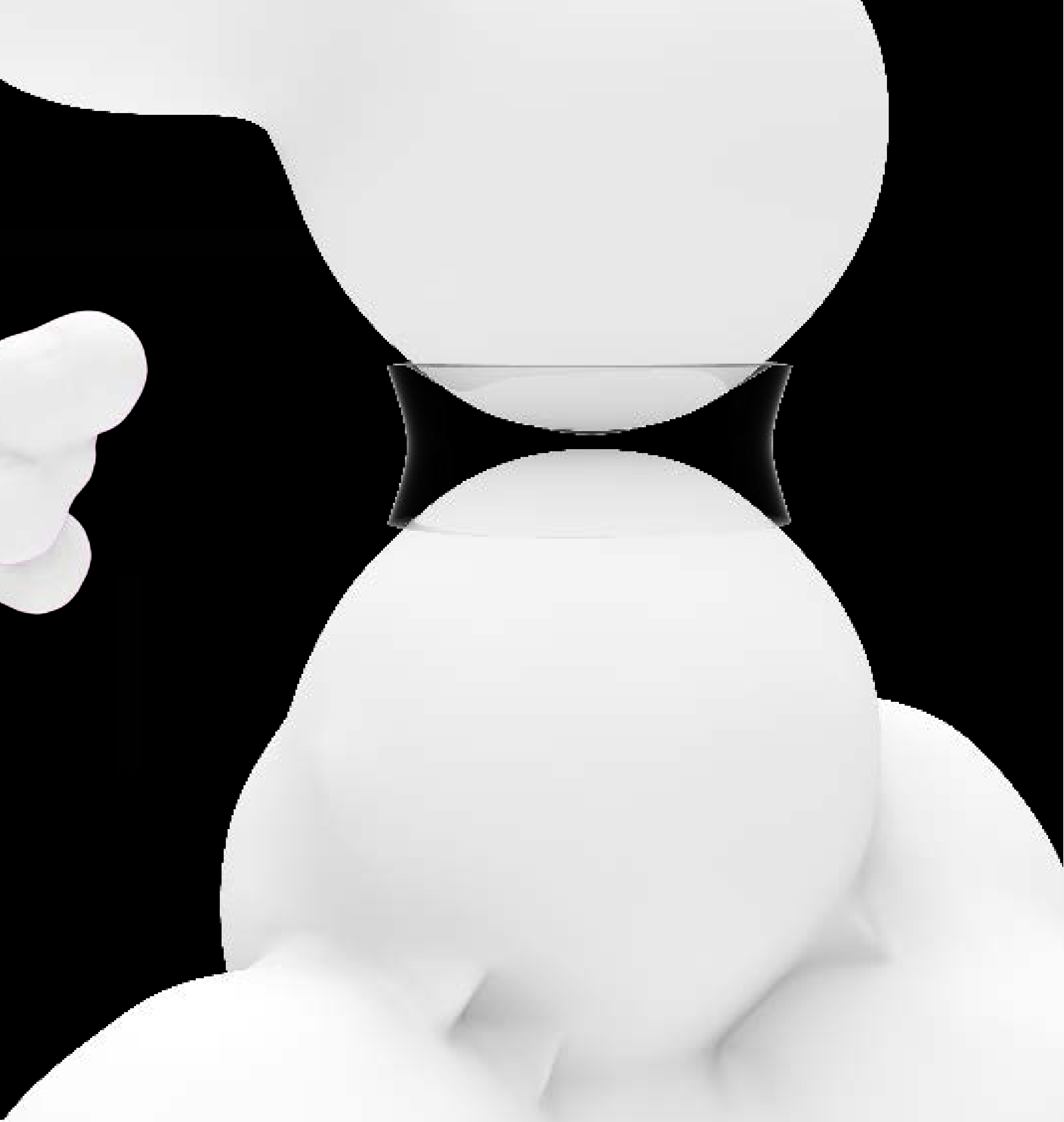
GEOMETRY DESIGN

Modular design : Cocoon 3D



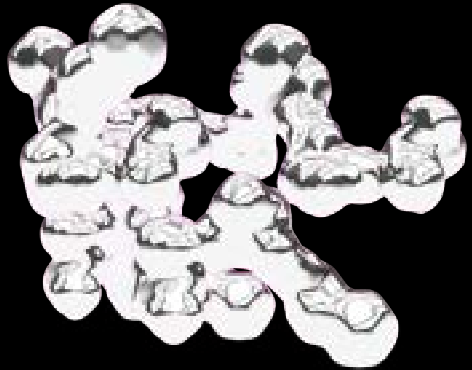
GEOMETRY DESIGN

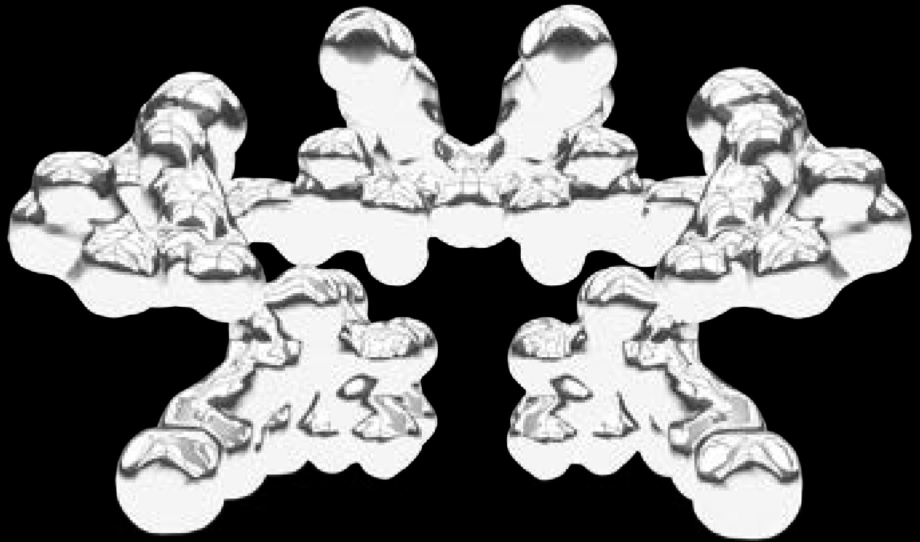
Jointing system

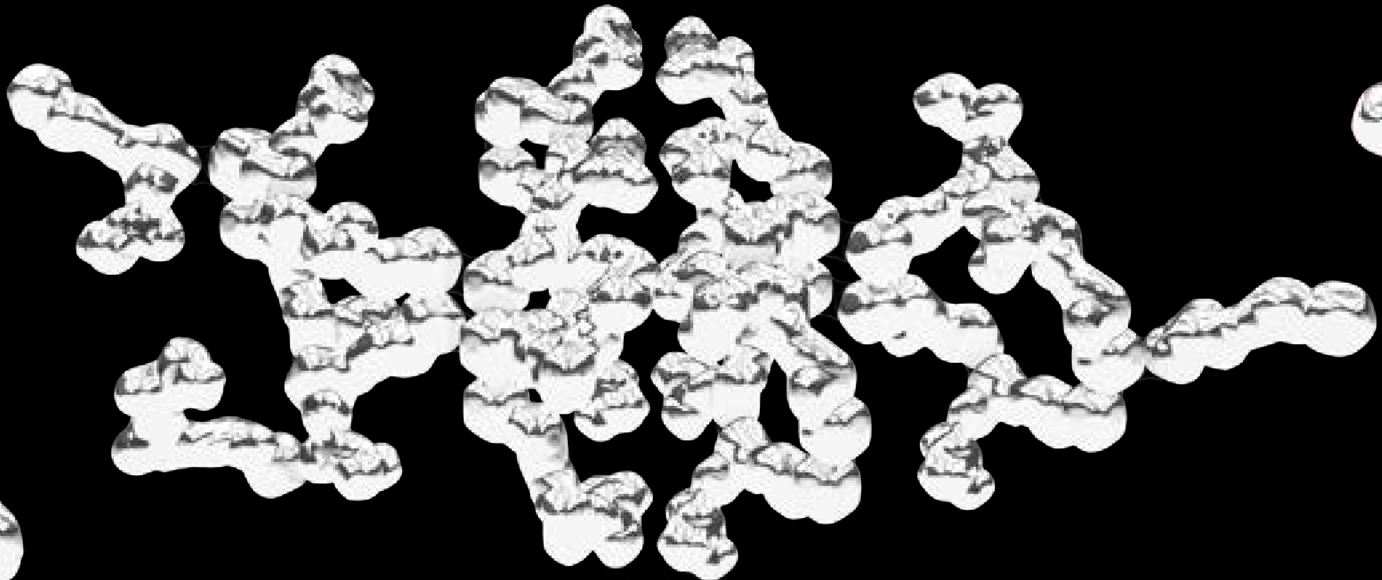


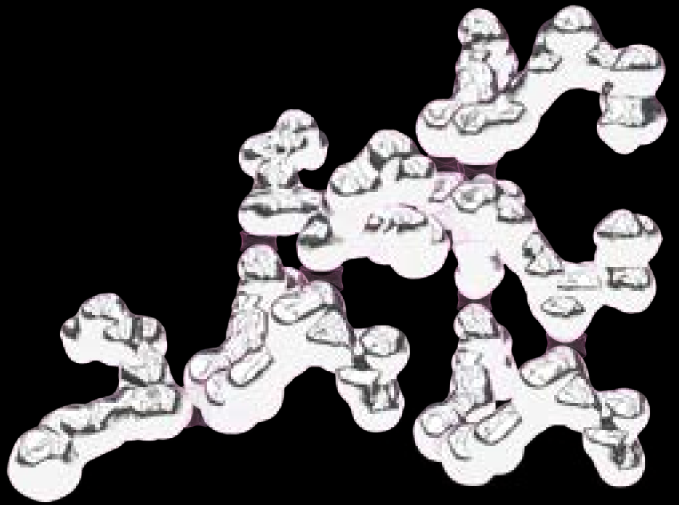
GEOMETRY DESIGN

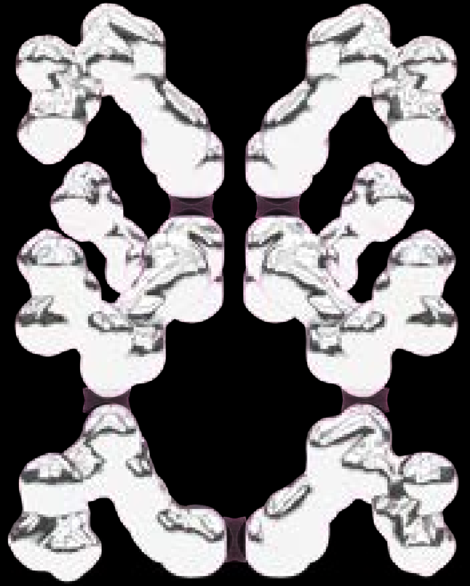
Aggregation











GEOMETRY DESIGN

Material

GEOMETRY DESIGN

Casting test : Coffee & plaster on 3D print mould

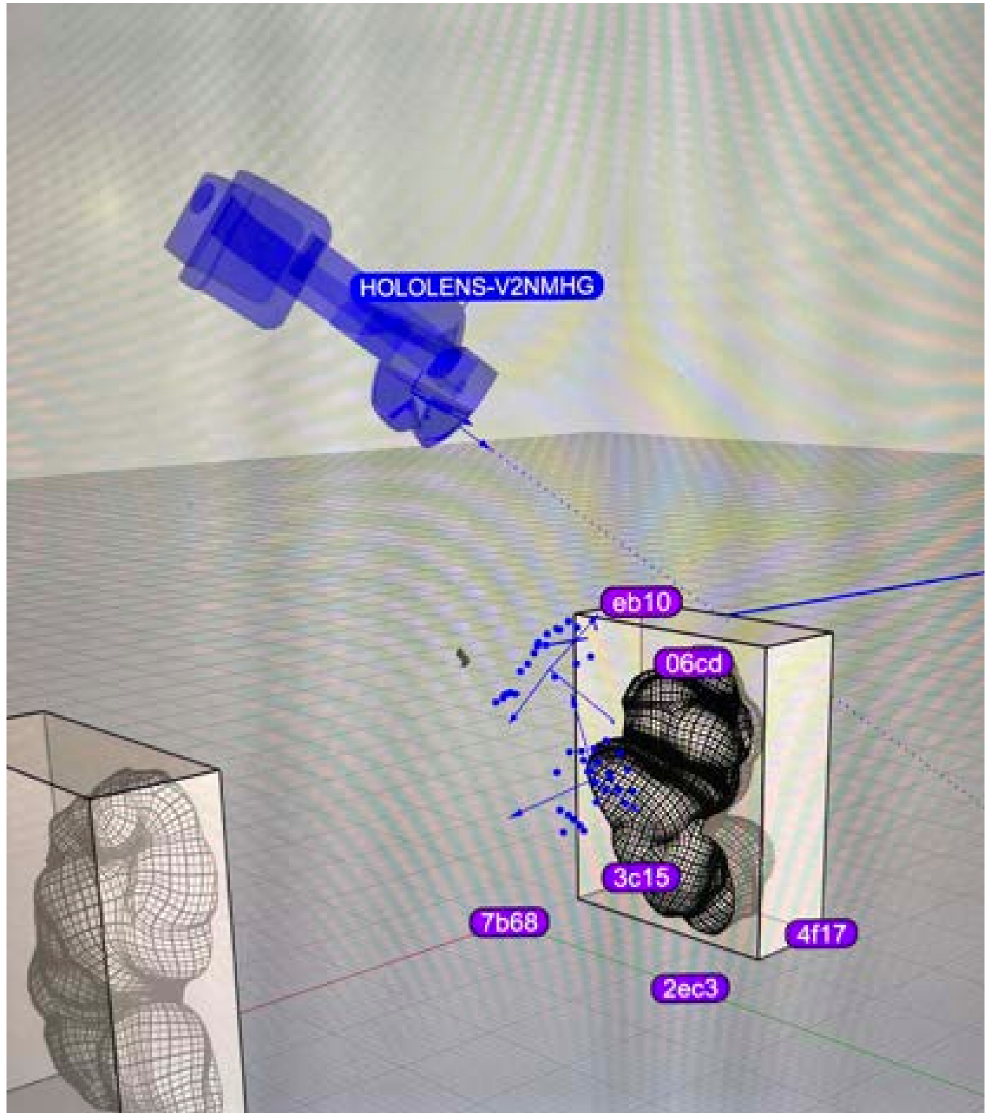
GEOMETRY DESIGN

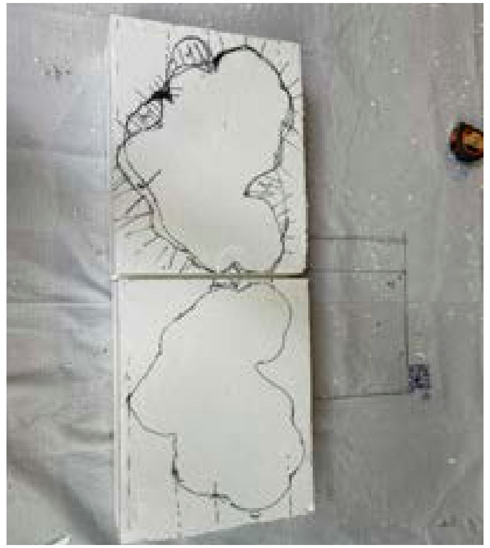
Carving mould, using Hololens as a guide line

Experimental design

Carving mould

CARVING **AR/MR**





CASTING TEST

Positive moulding

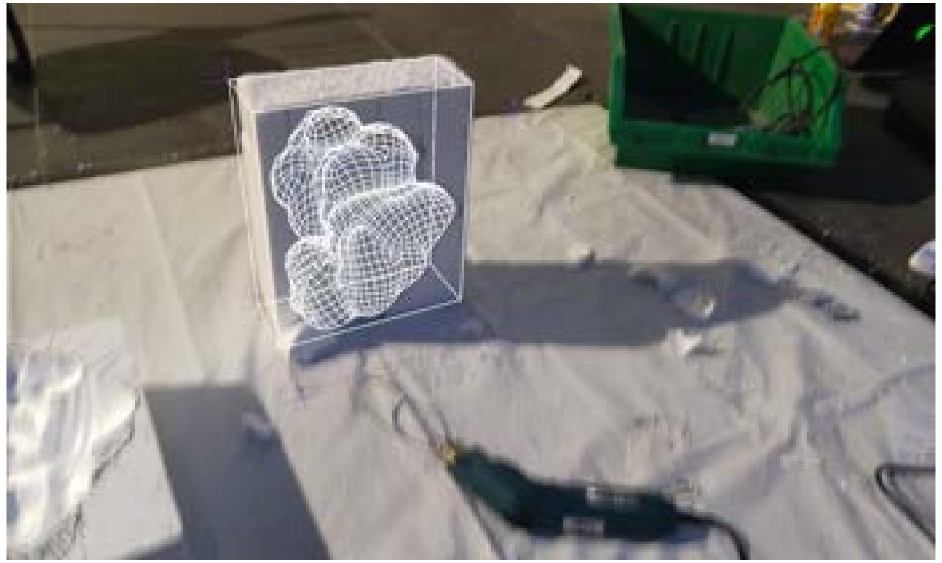
Base geometry (oil clay) --> Silicone mould (2 days curing) --> Plaster mould (2 hours) -->Material casting (1 day)

CARVING DETAILS (MR/AR)



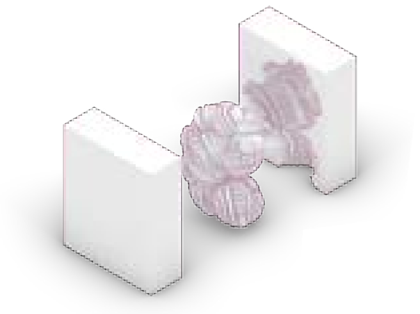






MOULDING & CASTING TEST

2-PART 3D PRINTING MOULD PROCESS



Base geometry

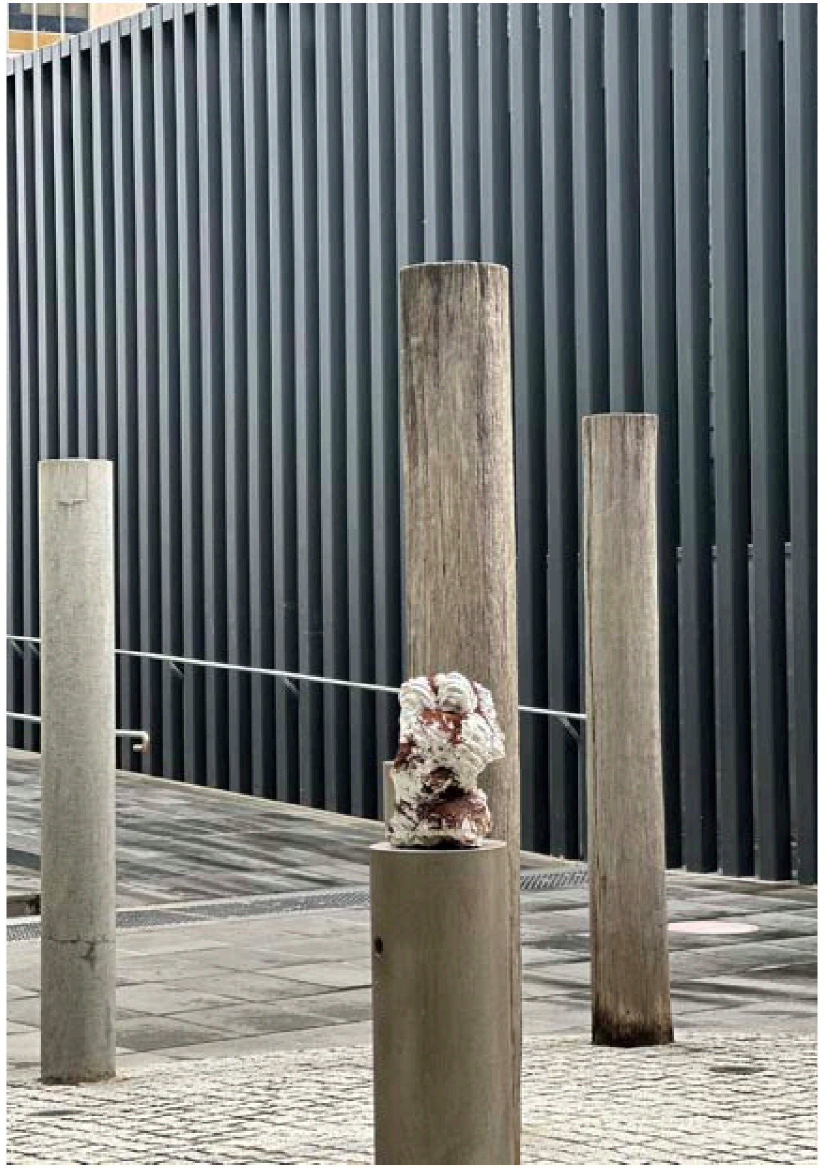
Plaster pouring

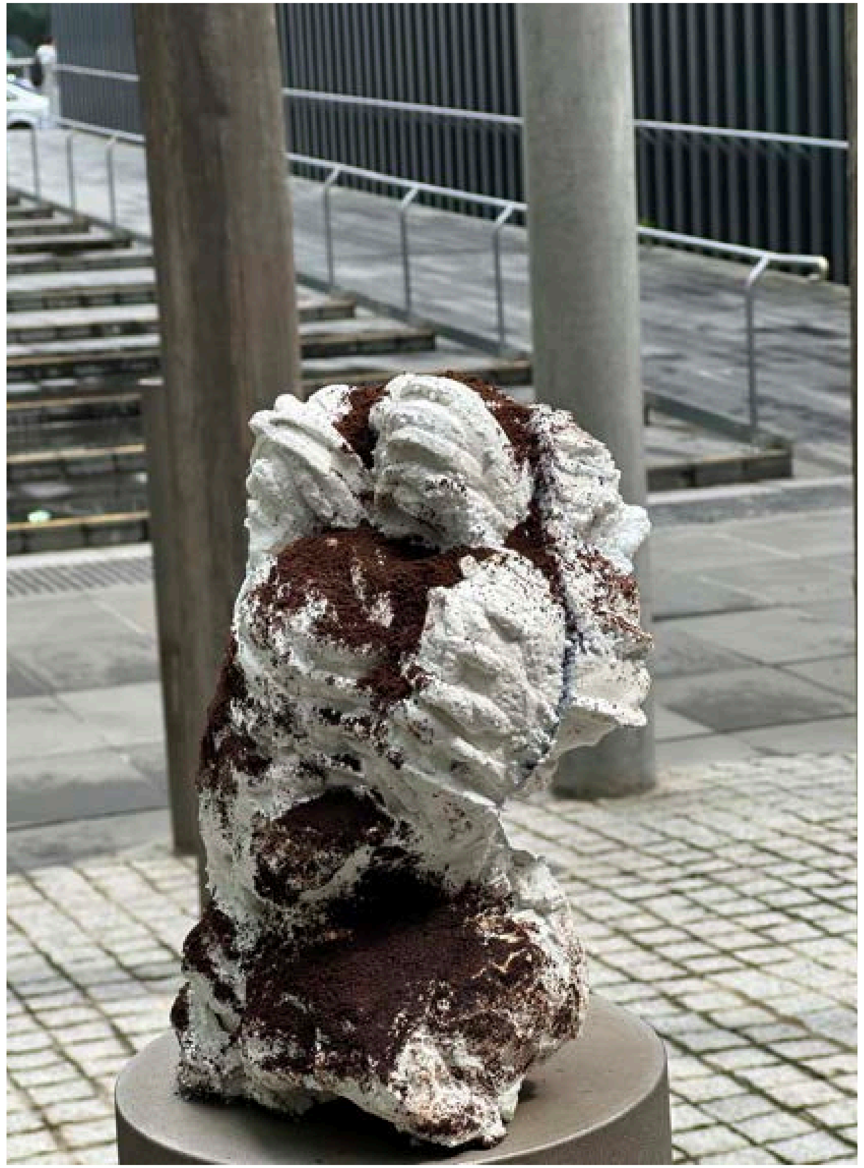
Casting&Demoulding

Final object

CASTED PIECE

CASTED PIECE





Reflection

This research investigates augmented fabrication workflows and sustainable biomaterial development through the use of spent coffee grounds as a composite material. Positioned within Melbourne's context — one of the highest coffee-consuming cities globally — the project explores how waste streams can be re-materialized into architectural form, transforming cultural identity and everyday consumption into spatial material expression. Coffee waste is employed not only as a sustainable resource, but as a medium that embodies local identity, circular material practice, and low-cost prototyping potential.

The study applies iterative material testing, focusing on binding strategies and dehydration processes. Early experimentation examined bio-adhesives including gelatin and cornstarch-based binders to align with ecological design principles. Despite initial failure to achieve structural cohesion, this phase revealed the mechanical limitations of biodegradable glues and highlighted the tension between ideal sustainability practices and functional performance. Subsequent testing identified polyurethane as the most suitable binding agent, enabling reliable composite integrity. Dehydration techniques — including sun-baking and heat toasting — further informed curing behavior and surface morphology.

Computational design and augmented fabrication tools supported the material workflow. Blob-like geometries were selected to respond to both material constraints and fabrication tool limits, emphasizing emergent form over fixed formal control. Casting was adopted as the primary technique due to its suitability for composite materials analogous to plaster. Mixed-reality visualization using Microsoft HoloLens played a critical role in bridging virtual and physical processes, enabling precise mold development and form interpretation within physical space.

This research demonstrates a hybrid design methodology that integrates computational form-finding, augmented fabrication, and circular material experimentation. Through iterative failures, recalibration, and tool-mediated feedback, the project reframes waste as a performative architectural medium and advances discourse on digital-material relations in sustainable design. The work contributes to emerging architectural research in localized resource reuse, mixed-reality craft, and computational fabrication strategies, forming a foundation for future exploration in zero-waste prototyping and bio-composite construction systems.